ISRO INTERVIEW PREPARATION

Agricultural Physics topics: concepts, flow charts and key papers

A brief revision guide built from the handwritten topic list (Prof. Sehgal's suggestions and course AP 607), covering agrometeorology, evapotranspiration, crop forecasting and insurance, soil and carbon, crop models, AI/ML and the basics.

43	~45	179
topic pages	flow charts and diagrams	papers and sources

Each topic page has: concept summary · flow chart · key points · interview one-liner · key papers. Figures marked 'schematic' or 'illustrative' are drawn for teaching. Verify scheme numbers against the latest official documents before quoting.

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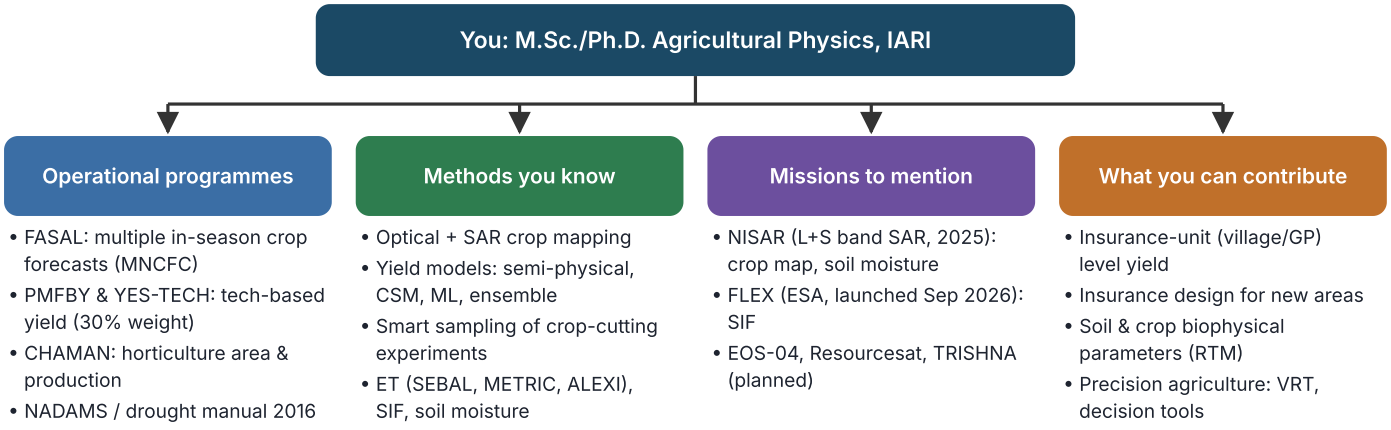
45 Rapid-fire interview questions with one-line answers

References

46 Key papers and sources

How to open the interview: your work in one map

Your notes plan a short introduction that links your IARI work to ISRO's programmes. Use the map below: start with what you studied, connect it to operational schemes, show the methods you know, then end with what you can contribute.



60-second structure

- **1.** Name, degree, division, thesis title in one line
- **2.** Core skill: remote sensing + crop/soil physics + modelling
- **3.** Link to ISRO: FASAL, YES-TECH, SAC's precision agriculture work
- **4.** One result or method you are proud of
- **5.** Close: 'I want to work on … (e.g., NISAR soil moisture, YES-TECH yield)'

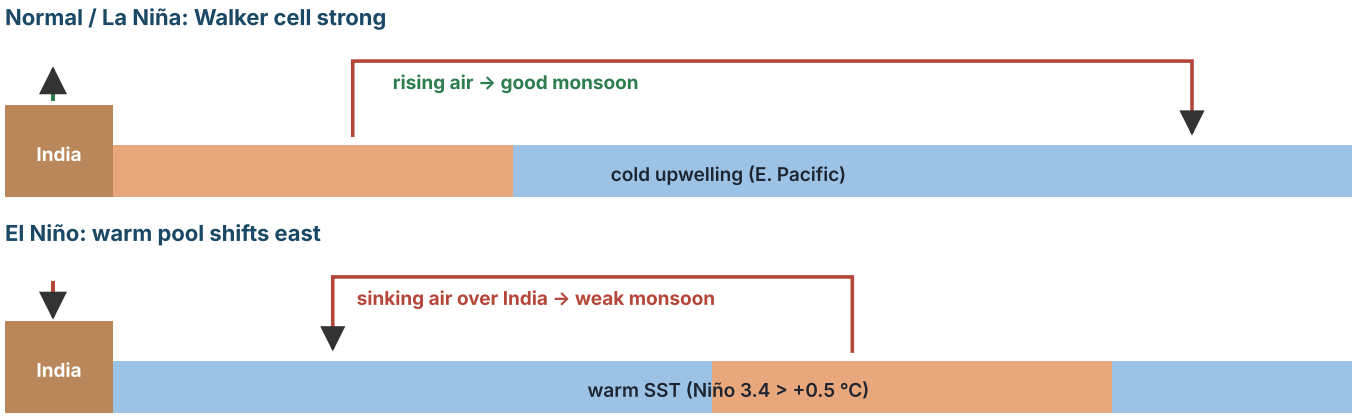
Likely opening questions

- Tell us about your research and its relevance to ISRO
- Which satellite data have you used and why?
- What is the difference between FASAL and YES-TECH?
- Which current mission excites you and why?
- How can remote sensing improve crop insurance?

Interview one-liner: My work sits where remote sensing meets crop and soil physics: estimating crop area, yield, ET and soil moisture from satellite data, which directly supports FASAL, YES-TECH under PMFBY and the new NISAR and FLEX missions.

Indian monsoon, ENSO, super El Niño and IOD

The south-west monsoon (June–September) gives about 75% of India's annual rainfall. Its strength depends strongly on sea surface temperatures in the Pacific (ENSO) and Indian Ocean (IOD). An El Niño shifts the Walker circulation east, so air sinks over India and the monsoon tends to weaken.



Schematic Walker circulation in normal and El Niño years

Term	Definition	Effect on India
El Niño	Niño 3.4 SST anomaly (ONI) ≥ +0.5 °C for 5 overlapping 3-month seasons	Tends to give below-normal monsoon (2002, 2009, 2015 droughts); not always (1997 normal)
La Niña	ONI ≤ −0.5 °C	Usually normal or excess monsoon
Super El Niño	ONI ≥ about +2 °C (1982–83, 1997–98, 2015–16)	Frequency projected to double with warming (Cai et al. 2014)
El Niño Modoki	Warming in central Pacific, cool on both sides	Stronger drying effect on India (Ashok et al. 2007)
Positive IOD	Warm western, cool eastern Indian Ocean (Saji et al. 1999)	Favours monsoon; can offset El Niño (1997, 2019)
MJO / MISO	Eastward-moving (30–60 day) and northward-moving intraseasonal oscillations	Active and break spells within the season

Monsoon basics

- Driven by land–sea thermal contrast and seasonal shift of the ITCZ
- Onset over Kerala ≈ 1 June; withdrawal from NW India from mid-September
- IMD long-period average (1971–2020) = 87 cm; normal = 96–104% of LPA
- Forecast: IMD two-stage long-range forecast (April, May/June), dynamical MMCFS + statistical ensemble

Why does an El Niño become strong?

- **Bjerknes feedback:** weaker trade winds → warm water moves east → smaller east–west SST gradient → trades weaken further
- **Westerly wind bursts** push warm water east (Kelvin waves)
- **Recharge oscillator:** large equatorial heat content beforehand (1997, 2015)
- Warming background raises odds of extreme events (Cai 2014)

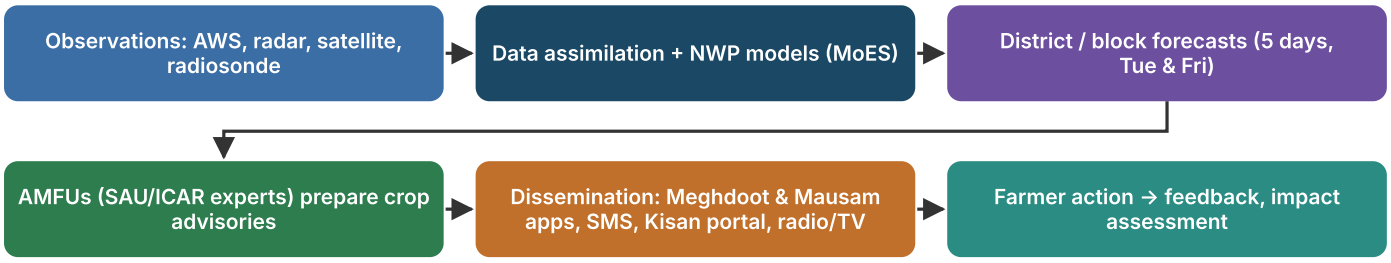
Interview one-liner: El Niño warms the central-eastern Pacific, shifts the Walker circulation's rising limb east and puts India under sinking air, so monsoons are often weak; a positive IOD can offset this, which is why 1997 had a normal monsoon despite a super El Niño.

Key papers / sources: Rasmusson & Carpenter (1983) Mon. Wea. Rev. 111:517 · Saji et al. (1999) Nature 401:360 (IOD) · Ashok et al. (2001) GRL 28:4499 · Kumar, Rajagopalan & Cane (1999) Science 284:2156 · Ashok et al. (2007) JGR 112:C11007 (Modoki) · Cai et al. (2014) Nature Climate Change 4:111

Weather forecasting and agromet advisory services in India

IMD (with IITM and NCMRWF under the Ministry of Earth Sciences) issues forecasts from nowcasts to seasonal outlooks. Forecasts are turned into crop-specific advice by the Agromet Advisory Service (Gramin Krishi Mausam Sewa, GKMS) through Agromet Field Units at SAUs and ICAR institutes.

Range	Lead time	Main tools	Agricultural use
Nowcast	0–3 h	Doppler weather radar, INSAT-3D/3DS, lightning network	Hail, thunderstorm, lightning (Damini app) warnings
Short range	1–3 days	High-resolution NWP (NCUM, WRF), Bharat Forecast System (~6 km)	Spraying, irrigation, harvest timing
Medium range	3–10 days	Global models (GFS T1534 ~12 km, NCUM), ensembles (GEFS, NEPS)	District/block agromet advisories (twice weekly)
Extended range	2–4 weeks	ERFS (CFS-based ensemble), MISO monitoring	Active/break spells, sowing windows, contingency
Long range	Month–season	MMCFS (coupled), statistical ensemble (SEFS), MME	Seasonal crop planning, drought preparedness



Agromet advisory chain (GKMS)

Advisory content

- Sowing/transplanting window, variety choice
- Irrigation scheduling (rain forecast), fertiliser timing
- Pest/disease alerts (humidity, temperature)
- Harvest and storage; heat/cold wave protection; livestock

Recent initiatives

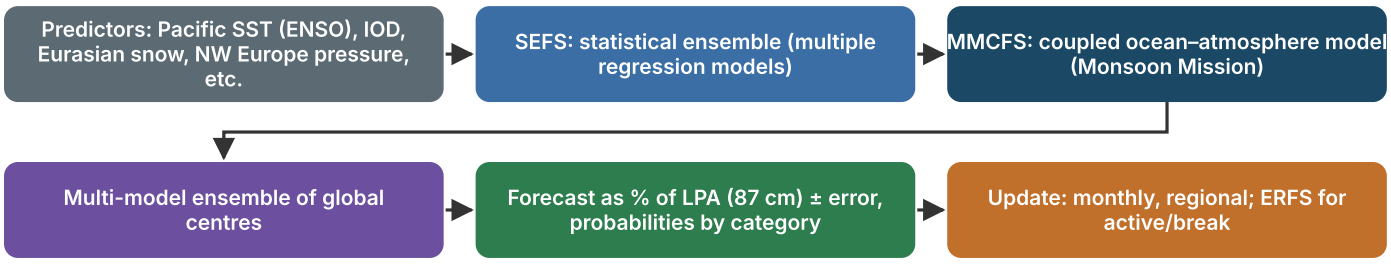
- Mission Mausam (Sept 2024): better observations, models, weather management
- Bharat Forecast System (IITM, 2025): ~6 km global model
- Block-level advisories; Krishi-DSS integration
- Impact: studies show advisories raise net returns

Interview one-liner: IMD runs forecasts from nowcasts to seasonal outlooks using radar, satellites and NWP models such as GFS, NCUM and ensembles; GKMS turns district and block 5-day forecasts into twice-weekly crop advisories through Agromet Field Units, delivered by Meghdoot, Mausam and SMS.

Key papers / sources: Rathore et al. (2001) IMD/NCMRWF agromet advisory · Venkatasubramanian et al. (2014) Mausam 65:365 (AAS) · MoES Mission Mausam (2024) · IMD Agromet Advisory Service Manual

Monsoon forecasting in India

IMD issues the south-west monsoon long-range forecast in two stages (mid-April and end-May/June) using a statistical ensemble forecast system and dynamical coupled models, plus forecasts of Kerala onset, monthly rainfall and extended-range active/break spells.



Long-range monsoon forecast system (two-stage)

Category	% of LPA
Deficient	< 90
Below normal	90–95
Normal	96–104
Above normal	105–110
Excess	> 110

Components

- **Stage 1 (April):** all-India seasonal rainfall forecast
- **Stage 2 (May/June):** update + regional and monthly forecasts
- **Kerala onset:** statistical model (±4 days); onset criteria: rainfall, wind depth, OLR
- **ERFS:** 2–4 week MISO-based outlook
- **Monsoon Mission (2012):** CFSv2-based dynamical model improvement (IITM)

Why forecasting is hard

- Monsoon shows large intraseasonal variability (active/break)
- ENSO–monsoon link weakened after 1980s; IOD, Eurasian snow matter
- Model biases in rainfall over India
- Predictability limited by chaotic internal variability

History

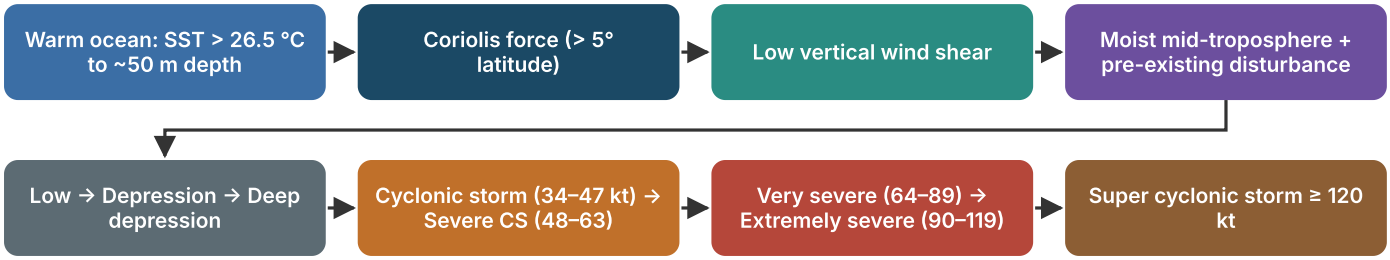
- Blanford (1884): Himalayan snow predictor; Walker (1920s): Southern Oscillation
- Power regression (Gowariker, 1989): 16 parameters
- 2003: 8- and 10-parameter models; 2007: ensemble statistical
- 2017→: dynamical MMCFS; 2021: new MME strategy

Interview one-liner: IMD forecasts the monsoon in two stages with a statistical ensemble of physical predictors and the dynamical Monsoon Mission coupled model, expressed as a percentage of the 87 cm LPA with category probabilities, plus onset, monthly and extended-range active/break forecasts.

Key papers / sources: Gowariker et al. (1989) Mausam 40:115 · Rajeevan et al. (2007) Clim. Dyn. 28:813 (ensemble statistical) · Rao et al. (2019) BAMS 100:2509 (Monsoon Mission) · Pai et al. (2017) IMD LRF documentation

Cyclones and weather hazards for agriculture

Tropical cyclones, heat and cold waves, frost, hail and floods are the main weather hazards for Indian agriculture (course AP 607, Weather Hazards and Management). Know the formation conditions, IMD categories and the criteria for heat and cold waves.



Cyclone formation conditions (top) and IMD intensity classes by maximum sustained wind (bottom)

Hazard	IMD criterion (plains, unless stated)	Agricultural impact and response
Heat wave	Tmax ≥ 40 °C and departure ≥ 4.5 °C (severe > 6.4 °C), or Tmax ≥ 45 °C (severe ≥ 47 °C); coastal ≥ 37 °C, hills ≥ 30 °C	Wheat terminal heat (grain filling), pollen sterility; irrigation, mulching, early sowing
Cold wave	Tmin ≤ 10 °C and departure ≤ −4.5 °C (severe ≤ −6.4 °C), or Tmin ≤ 4 °C	Chilling in mustard, potato, vegetables; light irrigation, smoke, covers
Frost	Leaf or ground temperature ≤ 0 °C on clear calm nights (radiative cooling, stable BL)	Ice formation in tissue; sprinkler irrigation, windbreaks
Cyclone	Bay of Bengal more active; peaks pre-monsoon (Apr–May) and post-monsoon (Oct–Dec)	Wind lodging, storm-surge salinity, flooding; early harvest advisories
Hail, flood	Convective storms (pre-monsoon, western disturbances); heavy rain ≥ 64.5 mm/day	Crop damage assessment with SAR and optical satellite data for insurance

Remote sensing role

- Cyclone tracking: INSAT-3D/3DR/3DS imagery, scatterometer winds
- Flood mapping: SAR (EOS-04, Sentinel-1) works through cloud
- Heat stress: LST from thermal bands; frost from night-time LST
- Damage for PMFBY localised calamities: before/after NDVI or SAR change

Climate trends to mention

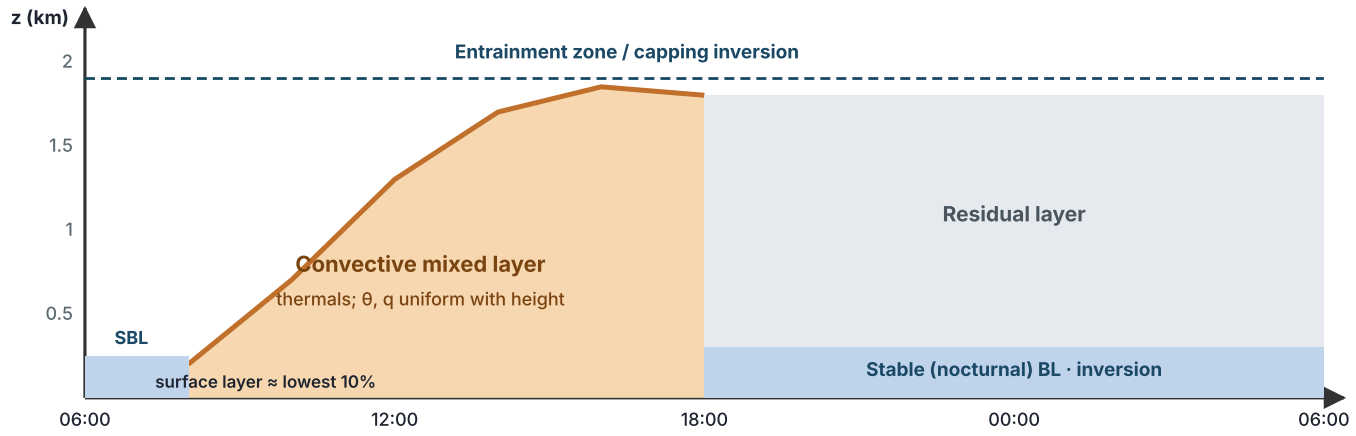
- Rising frequency of extremely severe cyclones in the Arabian Sea
- More frequent and longer heat waves (e.g., March 2022 wheat heat)
- Rapid intensification linked to warm ocean heat content
- Emanuel (2005): cyclone destructiveness rises with SST

Interview one-liner: A cyclone needs warm ocean above 26.5 °C, Coriolis force, low wind shear and a moist pre-existing disturbance; IMD grades it from depression to super cyclonic storm (≥ 120 knots), and SAR plus geostationary data are used to track it and map crop damage.

Key papers / sources: IMD terminology and criteria (heat/cold wave, cyclone classes) · Emanuel (2005) Nature 436:686 · Gray (1968) Mon. Wea. Rev. 96:669 (genesis parameters) · NDMA Heat Wave Guidelines (2019)

Atmospheric boundary layer: surface layer and PBL

The planetary boundary layer (PBL, about 100 m to 3 km) is the turbulent lowest part of the atmosphere that responds to the surface within about an hour. Its lowest tenth, the surface layer, has nearly constant fluxes and follows Monin–Obukhov similarity; this is where eddy covariance towers and ET models operate.



Diurnal evolution of the PBL (schematic)

Key equations

- Reynolds decomposition $u = \bar{u} + u'$; fluxes are covariances: $H = \rho c_p \cdot w'\theta'$, $LE = \rho L_v \cdot w'q'$, $\tau = -\rho \cdot u'w'$
- Friction velocity $u^* = \sqrt{(\tau/\rho)}$
- Log profile (neutral): $u(z) = (u^*/k) \cdot \ln[(z - d)/z_0]$, $k \approx 0.41$; crops: $d \approx 0.67h$, $z_0 \approx 0.1–0.12h$
- Obukhov length $L = -u^{*3}\theta_v/[k\, g\, (w'\theta_v')]$: $L < 0$ unstable, $L > 0$ stable; stability parameter z/L
- Aerodynamic resistance $r_a = \ln[(z_m - d)/z_{0m}] \cdot \ln[(z_h - d)/z_{0h}] / (k^2 u_z)$

Layers and terms

- **Surface layer:** lowest ~10% of PBL, constant-flux, Coriolis negligible
- **Roughness/canopy sublayer:** log law fails near elements
- **Mixed layer:** daytime convection; θ, q uniform
- **Entrainment zone / capping inversion:** PBL top
- **Residual layer** and **stable nocturnal BL** (50–300 m) at night
- **Ekman layer:** wind turns with height toward geostrophic
- Richardson number > ~0.25 → turbulence suppressed
- Dry adiabatic lapse rate 9.8 K km^{-1}

Why it matters

- ET models (Penman–Monteith, SEBAL, METRIC) use r_a, stability and blending height
- Frost and cold-air drainage under stable nights
- Residue-burning smoke trapped in shallow winter BL over the Indo-Gangetic Plain
- PBL schemes in weather and climate models

Measuring PBL height

- Radiosonde: inversion or θ jump
- Lidar, ceilometer: aerosol gradient at top
- Sodar, wind profiler
- Space lidar (CALIPSO) and reanalysis (ERA5)

Interview one-liner: The PBL is the turbulent lowest 100 m–3 km that responds to the surface; its bottom 10%, the surface layer, has near-constant fluxes and log profiles corrected by Monin–Obukhov stability, and this physics sets the aerodynamic resistance used in ET models and the dilution of pollutants.

Key papers / sources: Stull (1988) An Introduction to Boundary Layer Meteorology (Kluwer) · Monin & Obukhov (1954) Tr. Geofiz. Inst. AN SSSR 24:163 · Garratt (1992) The Atmospheric Boundary Layer (CUP) · Brutsaert (1982) Evaporation into the Atmosphere

Atmospheric stability and turbulence: Monin–Obukhov, Richardson, Reynolds

Stability decides whether vertical motion is enhanced (unstable), suppressed (stable) or unaffected (neutral). It controls turbulence, fluxes, frost, dispersion of smoke and aerodynamic resistance in ET models. These dimensionless numbers compare competing processes.

Quantity	Formula	Meaning / thresholds
Lapse rates	Environmental Γ vs dry adiabatic $\Gamma_d = g/c_p \approx 9.8$ K km^{-1}, saturated ≈ 4–7 K km^{-1}	$\Gamma > \Gamma_d$ absolutely unstable; $\Gamma < \Gamma_s$ absolutely stable; between = conditionally unstable
Potential temperature	$\theta = T(1000/p)^{0.286}$; virtual $\theta_v \approx \theta(1 + 0.61q)$	$d\theta/dz > 0$ stable, = 0 neutral, < 0 unstable
Brunt–Väisälä frequency	$N^2 = (g/\theta)(d\theta/dz)$	$N^2 > 0$ stable oscillation; $N^2 < 0$ convection
Reynolds number	$Re = UL/\nu$	Inertia vs viscosity; atmosphere Re ~10^7–10^8 → turbulent (pipe flow turbulent above ~2300)
Reynolds decomposition	$x = \bar{x} + x'$; flux = $\rho \cdot \text{mean}(w'x')$	Averaging rules: mean(x′) = 0; Reynolds stress $-\rho \cdot \text{mean}(u'w')$
Gradient Richardson number	$Ri = (g/\theta)(\partial\theta/\partial z) / (\partial u/\partial z)^2$	Buoyancy vs shear; Ri < 0 unstable; turbulence dies above $Ri_c \approx 0.25$
Flux / bulk Richardson	$R_f = (g/\theta)\text{mean}(w'\theta')/[\text{mean}(u'w')\partial u/\partial z]$; Ri_B from layer differences	Bulk Ri used to diagnose PBL height
Obukhov length	$L = -u^{*3}\theta_v / [k\, g\, \text{mean}(w'\theta_v')]$	L < 0 unstable, L > 0 stable, $\lvert L\rvert \to \infty$ neutral; $\lvert L\rvert \approx$ height where buoyancy = shear production
Stability parameter	$\zeta = z/L$; $\varphi_m = (1 - 16\zeta)^{-1/4}$ (unstable), $1 + 5\zeta$ (stable)	Businger–Dyer functions correct log profiles: Ψ_m, Ψ_h
Prandtl / Schmidt	$Pr = \nu/\kappa$; $Sc = \nu/D$; turbulent $Pr_t = K_m/K_h$	Ratios of momentum to heat/mass diffusivity
Pasquill classes	A (very unstable) … D (neutral) … F (stable)	From wind, insolation, cloud; used in smoke/pollutant dispersion

TKE budget: $\partial e/\partial t$ = shear production + buoyancy − dissipation (+ transport) → Daytime: buoyancy > 0 adds TKE → unstable, deep mixing → Night: buoyancy < 0 removes TKE → stable, shallow, frost & smog

Agricultural relevance

- Stability correction in SEBAL/METRIC H (Monin–Obukhov iteration)
- Frost nights: strong inversion, Ri large, calm air
- Spray drift and smoke dispersion depend on stability class
- Eddy covariance QC uses stationarity and ζ ranges

Interview checks

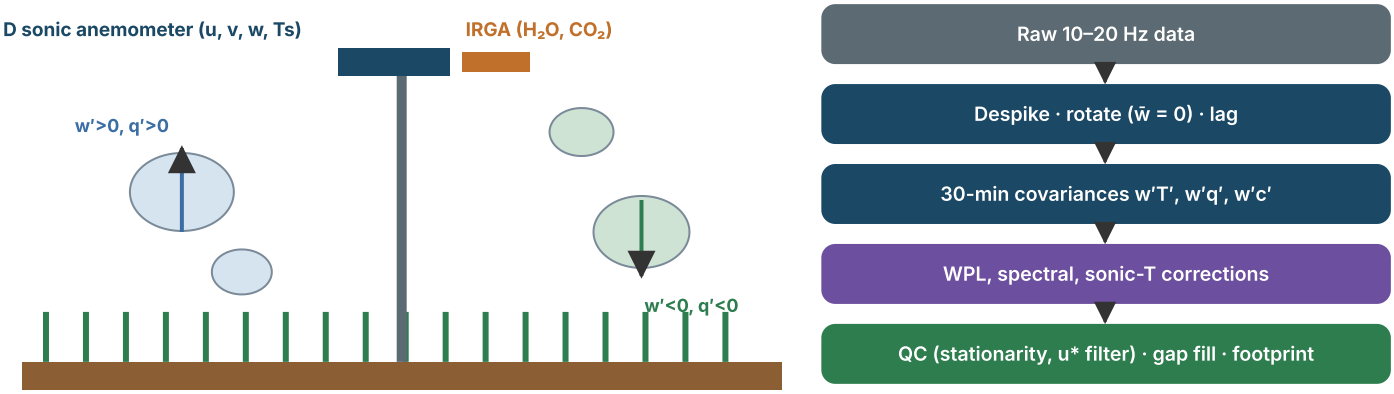
- Daytime crop canopy: usually unstable (L negative, −ζ ~ 0.1–1)
- Over irrigated field with warm advection: stable (H < 0)
- Neutral: windy overcast; log profile valid
- Ri and L both measure buoyancy vs shear; L is flux-based

Interview one-liner: Stability compares buoyancy with shear: the gradient Richardson number does it with gradients (turbulence stops above about 0.25), the Obukhov length with fluxes (negative when unstable), while the Reynolds number says the atmosphere is always turbulent and Reynolds decomposition turns that turbulence into measurable covariance fluxes.

Key papers / sources: Stull (1988) Boundary Layer Meteorology · Businger et al. (1971) J. Atmos. Sci. 28:181 · Dyer (1974) Bound.-Layer Meteorol. 7:363 · Monin & Obukhov (1954) · Pasquill (1961) Meteorol. Mag. 90:33

Micrometeorology: eddy covariance flux measurement

Eddy covariance measures the turbulent exchange of heat, water vapour and CO_2 directly. Rising and sinking eddies carry different amounts of vapour; the covariance of vertical wind (w′) and humidity (q′), sampled 10–20 times a second and averaged over 30 minutes, is the flux.



Eddies carry moist air up and drier air down; processing chain on the right

$$\lambda E = \lambda \cdot \rho_a \cdot \mathrm{cov}(w', q') \cdot H = \rho_a c_p \cdot \mathrm{cov}(w', T') \cdot F_c = \mathrm{cov}(w', c') \cdot NEE = F_c + \text{storage}$$

Set-up and assumptions

- 3-D sonic anemometer + infrared gas analyser (open or closed path)
- Height ≈ 2× canopy; upwind fetch ≈ 100× height
- Flat, homogeneous surface; stationary turbulence; $\bar{w}$ = 0 after rotation
- Footprint: upwind source area grows with height and stability

Energy balance closure

- (H + λE)/(Rn − G) typically 0.7–0.9
- Causes: large eddies, advection, storage, footprint mismatch
- Fix: Bowen-ratio-preserving forced closure
- Night-time: low u* → underestimates; filtered and gap-filled

Uses for ISRO

- Ground truth for satellite ET (SEBAL, MOD16), GPP and SIF
- Carbon budgets: NEE, GPP, Reco partitioning
- ISRO-GBP National Carbon Project flux towers over forests and mangroves
- FLUXNET/AsiaFlux networks

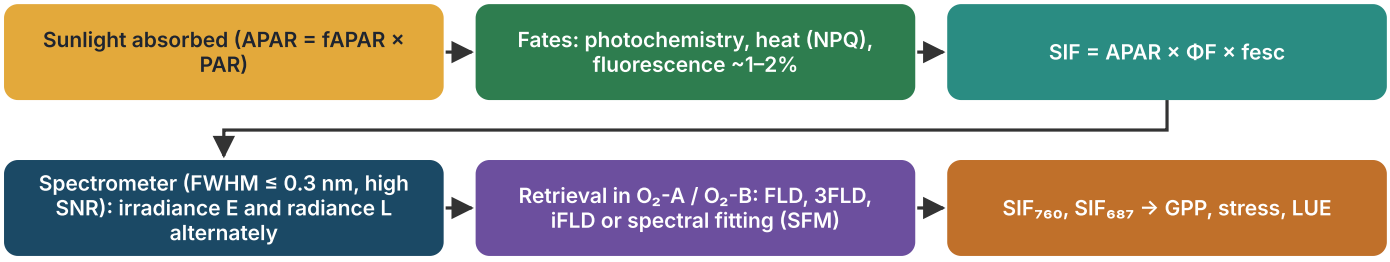
Other ET methods	Principle	Scale
Weighing lysimeter	Mass change of soil block; only direct measure	Point (m²)
Bowen ratio (BREB)	β = γΔT/Δe; λE = (Rn − G)/(1 + β)	Field
Scintillometer (LAS)	Refractive-index fluctuations → H over 0.5–5 km	Satellite pixel
Sap flow	Heat pulse in stems; transpiration only	Plant

Interview one-liner: Eddy covariance computes fluxes as the 30-minute covariance of vertical wind with temperature, humidity or CO_2 measured at 10–20 Hz by a sonic anemometer and gas analyser; it needs flat homogeneous fetch, WPL and rotation corrections, and usually closes the energy balance only to 70–90%.

Key papers / sources: Baldocchi (2003) Global Change Biology 9:479 · Webb, Pearman & Leuning (1980) QJRMS 106:85 · Aubinet, Vesala & Papale (2012) Eddy Covariance: A Practical Guide (Springer) · Foken (2008) Ecol. Appl. 18:1351 (closure)

Sun-induced fluorescence: ground measurement and FLEX

Chlorophyll re-emits 1–2% of absorbed light as fluorescence (650–850 nm; peaks at 685 and 740 nm). Because it comes straight from photosystem II, SIF tracks photosynthesis better than greenness indices. It is too weak to see in ordinary bands, so it is retrieved inside dark atmospheric absorption lines (O_2-B at 687 nm, O_2-A at 760 nm).



From absorbed light to SIF, and the ground measurement chain

FLD: $F = (E_{out} \cdot L_{in} - E_{in} \cdot L_{out}) / (E_{out} - E_{in})$ (in = inside the absorption line, out = shoulder)

Ground and airborne systems

- Tower systems: FloX, PhotoSpec, automated QE-Pro spectrometers (continuous)
- Field spectroradiometers with sub-nm resolution
- UAV and airborne: HyPlant (FLEX demonstrator)
- Needs accurate irradiance, dark current and O_2 path correction

FLEX mission (ESA Earth Explorer 8)

- Launched 15 Sep 2026 on Vega-C with Sentinel-3C
- FLORIS imaging spectrometer, ~500–780 nm, 300 m pixels
- Flies in tandem with Sentinel-3 (OLCI, SLSTR) for reflectance and LST
- Full fluorescence spectrum, both peaks; vegetation stress and GPP

Other satellite SIF

- GOSAT (Frankenberg 2011), GOME-2, OCO-2 (2014), TROPOMI (2017)
- SIF detects drought stress before NDVI drops
- Links to your Chl fluorescence/PAM knowledge (Fv/Fm, NPQ)

Interview one-liner: SIF is the 1–2% of absorbed light re-emitted by chlorophyll; it is retrieved from the infilling of the O_2-A and O_2-B absorption lines with FLD or spectral fitting from high-resolution spectrometers, and ESA's FLEX (launched September 2026) now maps it globally at 300 m.

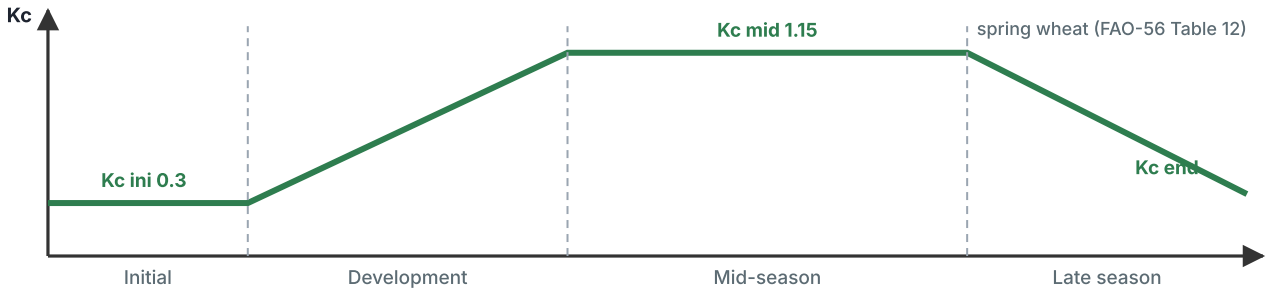
Key papers / sources: Meroni et al. (2009) RSE 113:2037 · Frankenberg et al. (2011) GRL 38:L17706 · Porcar-Castell et al. (2014) J. Exp. Bot. 65:4065 · Cogliati et al. (2015) RSE 164:270 · Drusch et al. (2017) IEEE TGRS 55:1273 (FLEX) · Mohammed et al. (2019) RSE 231:111177

Evapotranspiration (1): FAO-56 classical approach

FAO-56 (Allen et al. 1998) is the standard for crop water requirement. Reference ET (ET0) from weather data describes atmospheric demand for a well-watered grass; a crop coefficient Kc converts it to crop ET, and a stress coefficient Ks reduces it to actual ET when soil water runs short.

$$ET_0 = [0.408\Delta(R_n - G) + \gamma \cdot 900/(T + 273) \cdot u_2(e_s - e_a)] / [\Delta + \gamma(1 + 0.34u_2)]$$

Weather: Tmax, Tmin, RH/Tdew, u_2, sunshine/Rs → Reference ET0 (grass 0.12 m, rs 70 s/m, albedo 0.23) → × Kc (crop stage) → ETc → × Ks (root-zone water balance) → ETa



Single crop coefficient curve

Crop coefficients

- ETc = Kc × ET0; Kc follows initial, development, mid and late stages
- Dual: ETc = (Ks·Kcb + Ke) × ET0 separates transpiration (Kcb) and soil evaporation (Ke)
- Kc mid/end adjusted for wind and RH_{min}
- Typical Kc mid: wheat 1.15, rice 1.20, maize 1.20, sugarcane 1.25

Water stress Ks

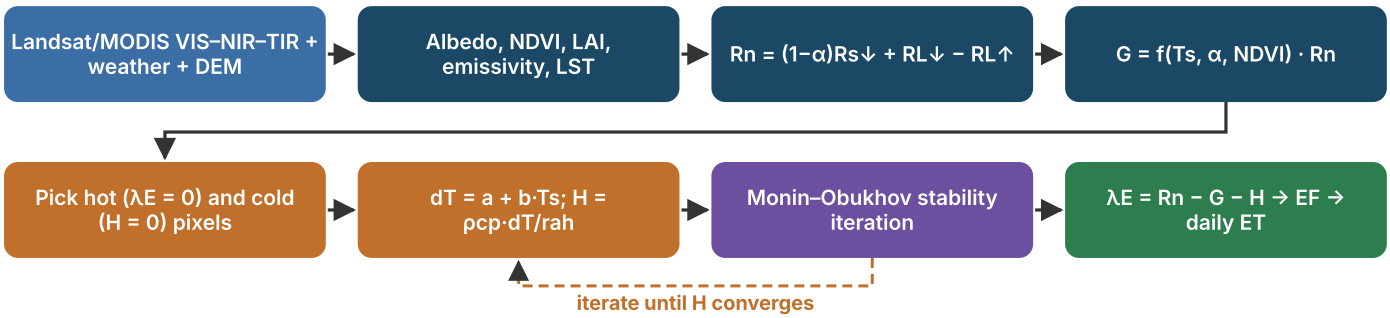
- $TAW = 1000(\theta_{FC} - \theta_{WP})Z_r$; $RAW = p \cdot TAW$ ($p \approx 0.5$)
- $Ks = (TAW - D_r)/[(1 - p)TAW]$ once depletion $D_r >$ RAW
- Irrigate when D_r reaches RAW
- AET/ET0 = actual Kc: the ratio satellites can map

Interview one-liner: FAO-56 computes reference ET0 from weather with the Penman–Monteith equation for a standard grass, multiplies by a stage-dependent Kc for crop ET and by Ks from a soil water balance for actual ET; satellites estimate actual ET directly, and AET/ET0 gives an actual crop coefficient.

Key papers / sources: Allen, Pereira, Raes & Smith (1998) FAO Irrigation & Drainage Paper 56 · Monteith (1965) Symp. Soc. Exp. Biol. 19:205 · Doorenbos & Pruitt (1977) FAO-24

Evapotranspiration (2): satellite surface energy balance

Satellites cannot see water vapour leaving a field, but thermal bands show its cooling effect. Energy-balance models compute net radiation, soil heat flux and sensible heat from albedo, NDVI and land surface temperature, and take latent heat (ET) as the remainder: λE = Rn − G − H.



SEBAL/METRIC workflow

Model	Type	Key idea	Note
SEBAL (Bastiaanssen 1998)	Single-source	Hot/cold pixels calibrate dT–Ts line inside each image; minimal ground data	Flat terrain; pixel choice subjective
METRIC (Allen 2007)	Single-source	Cold pixel ET = 1.05 × alfalfa ETr; ETrF for daily/seasonal; DEM slope correction	Needs hourly weather station
S-SEBI (Roerink 2000)	Contextual	Ts–albedo dry and wet edges; EF = $(T_H - Ts)/(T_H - T_{\lambda E})$	Used by SAC with Kalpana-1
SEBS (Su 2002)	Single-source	Physical dry and wet limits per pixel; dynamic kB^{-1}	Many inputs
ALEXI (Anderson 1997)	Two-source	Morning LST rise from geostationary + boundary-layer growth; no air temperature needed; separates E and T	5–10 km; DisALEXI sharpens to 30 m
TSEB (Norman 1995)	Two-source	Soil and canopy temperatures; Priestley–Taylor first guess	Sparse and row crops

Why thermal?

- Evaporating crop stays cool; stressed crop heats up
- 1 K error in absolute dT ≈ tens of W m^{-2} in H; hot/cold calibration cancels most error
- TIR sensors: Landsat 100 m, MODIS 1 km, INSAT-3D 4 km, ECOSTRESS 70 m, TRISHNA ~57 m (planned)

Single vs dual source

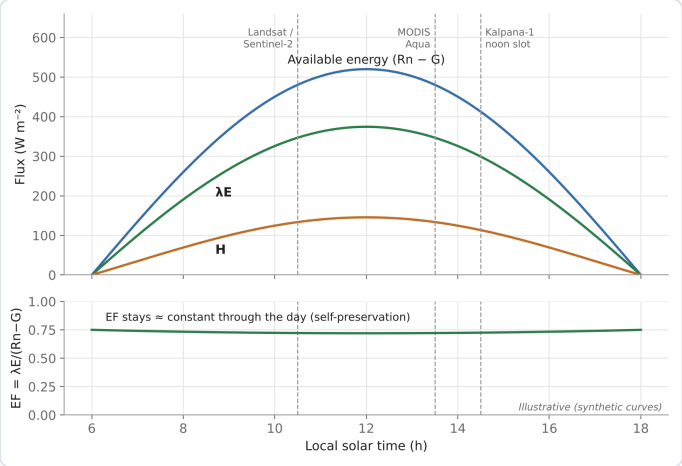
- Single-source: one surface temperature; fine for dense crops
- Dual-source (TSEB, ALEXI): soil and canopy separately → partial cover, E/T partition
- Validation: eddy covariance, lysimeter, scintillometer

Interview one-liner: SEBAL and METRIC are single-source residual models that calibrate sensible heat with a hot and a cold pixel; ALEXI is a two-source model that uses the morning rise in surface temperature from geostationary satellites and separates soil evaporation from transpiration.

Key papers / sources: Bastiaanssen et al. (1998) J. Hydrol. 212–213:198 · Allen, Tasumi & Trezza (2007) J. Irrig. Drain. Eng. 133:380 · Anderson et al. (1997) RSE 60:195 · Norman, Kustas & Humes (1995) Agric. For. Meteorol. 77:263 · Su (2002) HESS 6:85 · Li et al. (2009) Sensors 9:3801 · Bhattacharya et al. (2010) J. Hydrol. 387:65

Evapotranspiration (3): from instantaneous to daily ET

A polar satellite sees a field once (≈10:30 for Landsat, 13:30 for MODIS-Aqua). Daily ET is obtained by assuming a ratio stays constant through the day: evaporative fraction (EF = λE/(Rn − G)) or reference ET fraction (ETrF = ET/ETr, i.e. AET/ET0-type). Your notes: 'the ratio doesn't change, so it is used to calculate 24-h ET'.



Illustrative clear-sky day: fluxes vary strongly but EF stays nearly constant (self-preservation)

Method	Formula	Used in
Constant EF (Sugita & Brutsaert 1991)	$\lambda E_{24} = EF_{inst} \times Rn_{24}$ ($G_{24} \approx 0$); ALEXI × 1.1 for night ET	SEBAL, SEBS, S-SEBI, triangle
Constant ETrF (Allen 2007)	$ET_{24} = (ET_{inst}/ETr_{inst}) \times ETr_{24}$	METRIC; handles afternoon advection
Sine function (Jackson 1983)	$ET_d/ET_i = 2N / [\pi \sin(\pi t/N)]$	Clear days
Geostationary	$\Lambda_d \approx \Lambda_{noon}$; $Q_d = 2Q_{noon}/[\pi \sin(\pi(t_{sat}-t_{rise})/(t_{set}-t_{rise}))]$	Kalpana-1 (SAC)

Worked example (one wheat pixel)

- Overpass: Rn 600, G 60, H 150 → λE = 390 W m^{-2}; EF = 0.72
- ET_{inst} = 390 × 3600 / 2.45×10^6 = 0.57 mm h^{-1}
- EF method: 0.72 × 13.5 MJ m^{-2} d^{-1} / 2.45 = **4.0 mm d^{-1}**
- ETrF method: (0.57/0.75) × 6.0 = **4.6 mm d^{-1}**
- Sine (t = 5 h, N = 12 h): 0.57 × 7.9 = **4.5 mm d^{-1}**

Why it works and when it fails

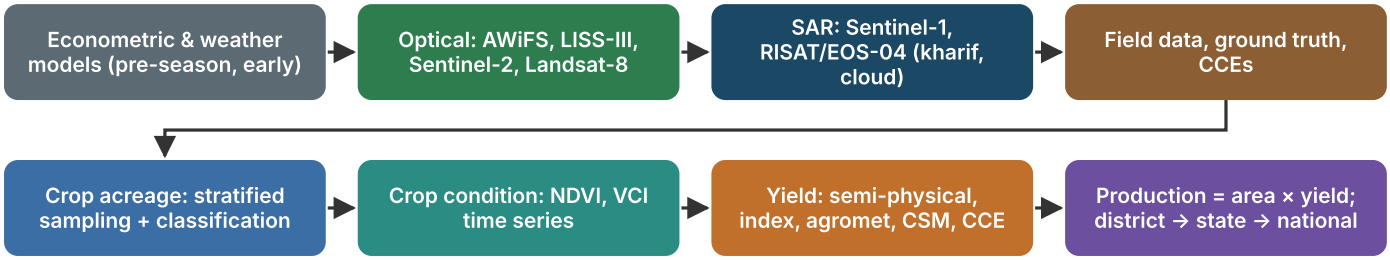
- Partitioning set by slow variables (soil moisture, LAI, resistances); Rn and Ts change fast
- Midday EF slightly underestimates (EF is concave over the day)
- Night ET ≈ 10% of daily
- Breaks with clouds and afternoon advection (use ETrF)
- Seasonal: interpolate EF/ETrF between images × daily ET0

Interview one-liner: Daily ET is the instantaneous EF (or ETrF) multiplied by daily available energy (or daily reference ET), because on clear days the share of energy used for ET stays nearly constant while radiation varies.

Key papers / sources: Sugita & Brutsaert (1991) WRR 27:747 · Crago (1996) J. Hydrol. 180:173 · Jackson et al. (1983) Agric. Water Manage. 7:351 · Allen et al. (2007) · Li et al. (2009) Sensors 9:3801

FASAL: multiple in-season crop production forecasts

FASAL (Forecasting Agricultural output using Space, Agro-meteorology and Land based observations) extended ISRO's earlier CAPE project to multiple in-season national forecasts. Early in the season the crop is too small to see, so econometric and weather models lead; as the canopy develops, remote sensing takes over.



FASAL logic (methods shown are the general approach)

Agency	Role
MNCFC, New Delhi (est. April 2012)	Operational forecasts at district, state, national level for 9 crops
SAC-ISRO, Ahmedabad	Developed RS methodologies (8 crops transferred to MNCFC)
NRSC-ISRO, Hyderabad	Satellite data supply
IMD	Agromet (correlation-weighted regression) and crop simulation yield models
IEG, Delhi	Econometric forecasts
State RS centres, SAUs, state depts	Ground truth and validation

Crops and forecasts

- Jute, kharif rice, sugarcane, cotton, rapeseed-mustard, rabi sorghum, wheat, rabi pulses, rabi rice
- 2017–18: 17 forecasts for 9 crops
- Early forecast: long lead, low accuracy; pre-harvest: high accuracy, short lead

Yield models used

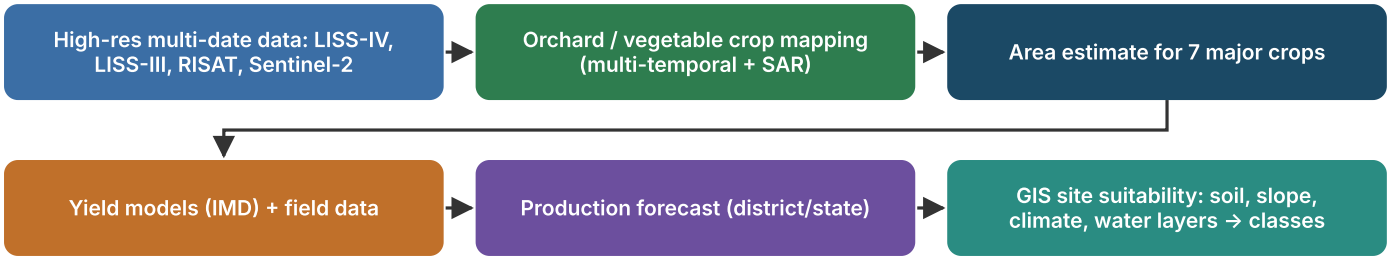
- **Semi-physical:** biomass = Σ fAPAR × PAR × RUE × stress; yield = biomass × HI
- **Index-based:** yield regressed on peak NDVI/VCI
- **Agromet:** yield vs weather by phenophase (IMD)
- **Crop simulation:** WOFOST/InfoCrop-type
- **CCE:** ground truth

Interview one-liner: FASAL integrates econometric, weather, optical and SAR satellite and field observations to give multiple in-season forecasts of crop area, yield and production; MNCFC runs it operationally using SAC's methods, with SAR essential for cloudy kharif seasons.

Key papers / sources: Parihar & Oza (2006) Proc. SPIE 6411 (FASAL concept) · Dadhwal et al. (2002) Acta Astronautica 51:533 (CAPE) · Ray et al. (2014) ISPRS Archives XL-8:1349 · MNCFC / DES, Ministry of Agriculture (FASAL pages)

CHAMAN: horticulture assessment with geoinformatics

CHAMAN (Coordinated Horticulture Assessment and Management using geoiNformatics) was launched in September 2014 under the Mission for Integrated Development of Horticulture (MIDH). It uses satellite data to map horticultural crops, forecast production and plan area expansion through site suitability.



CHAMAN components (general method)

Aspect	FASAL	CHAMAN
Crops	Field crops (wheat, rice, jute, cotton ...)	Horticulture (mango, banana, citrus, potato, onion, tomato ...)
Aim	Multiple in-season production forecasts	Area & production + development planning
Data	AWiFS, LISS-III, Sentinel-1/2, Landsat-8	LISS-IV, LISS-III, RISAT, high resolution
Resolution need	Medium (large contiguous fields)	High (small fragmented orchards)
Coordinator	MNCFC	MNCFC with SAC, NRSC, NESAC

Why horticulture is harder

- Small, fragmented orchards and intercropping
- Perennial canopies change little through the season
- Similar spectra across fruit species; age and variety effects
- Needs high-resolution, multi-temporal optical and SAR

Scope and examples

- Phase I: 12 states, 185 districts, 7 crops; Phase II: 15 states, 265 districts
- MRSAC: banana, mango, citrus orchard inventory; onion/tomato with RISAT
- Assam: site suitability for banana, Assam lemon, turmeric, pineapple
- Related: NADAMS drought assessment at MNCFC

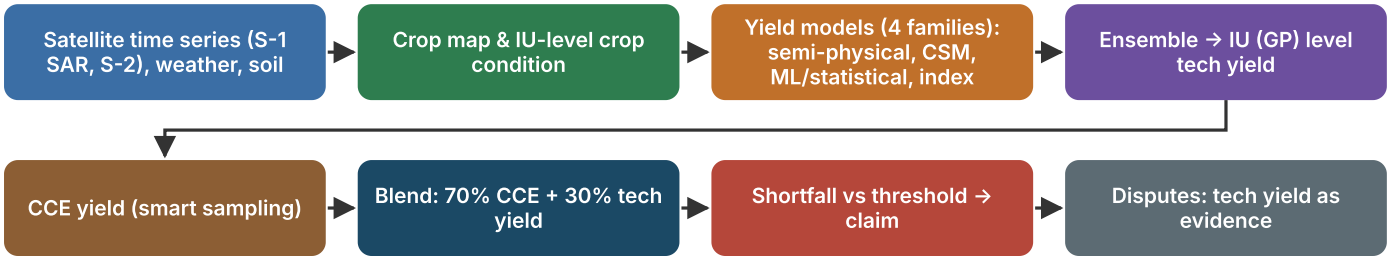
Interview one-liner: CHAMAN, launched in 2014 under MIDH, uses high-resolution optical and SAR data to map and forecast seven major horticultural crops and GIS overlays for site suitability; it needs finer data than FASAL because orchards are small, perennial and spectrally similar.

Key papers / sources: Ministry of Agriculture (MIDH) CHAMAN documents · NNRMS Bulletin (ISRO, May 2016) · MRSAC CHAMAN project reports · Ray et al. (2016) on horticulture assessment (MNCFC)

PMFBY and YES-TECH: technology for crop insurance

PMFBY (2016) pays claims when the actual yield of a notified insurance unit (IU: village/gram panchayat for major crops) falls below the threshold yield. Actual yield traditionally comes from crop-cutting experiments (CCEs); YES-TECH (2023) adds technology-based yield with mandatory 30% weight for paddy and wheat from Kharif 2023.

Claim = (Threshold yield − Actual yield) / Threshold yield × Sum insured; Threshold = avg. of best 5 of last 7 years × indemnity level



YES-TECH workflow (your notes: '4 + 1 → ensemble', IU/panchayat level)

PMFBY features

- Premium (farmer): 2% kharif, 1.5% rabi, 5% commercial/horticulture
- Area approach for widespread loss; individual assessment for localised calamities and post-harvest loss
- Tech: YES-TECH (yield), WINDS (weather network), CROPIC (crop photos)
- Restructured Weather Based Crop Insurance (RWBCIS) for horticulture

YES-TECH points

- Manual released 2023 (MoA&FW with MNCFC)
- Blend 70:30 (CCE : tech) for paddy and wheat; some states (MP) use 100% for selected crops
- Models must be validated against CCEs before acceptance
- Reduces CCE burden, delays and disputes

Horticulture / weather side

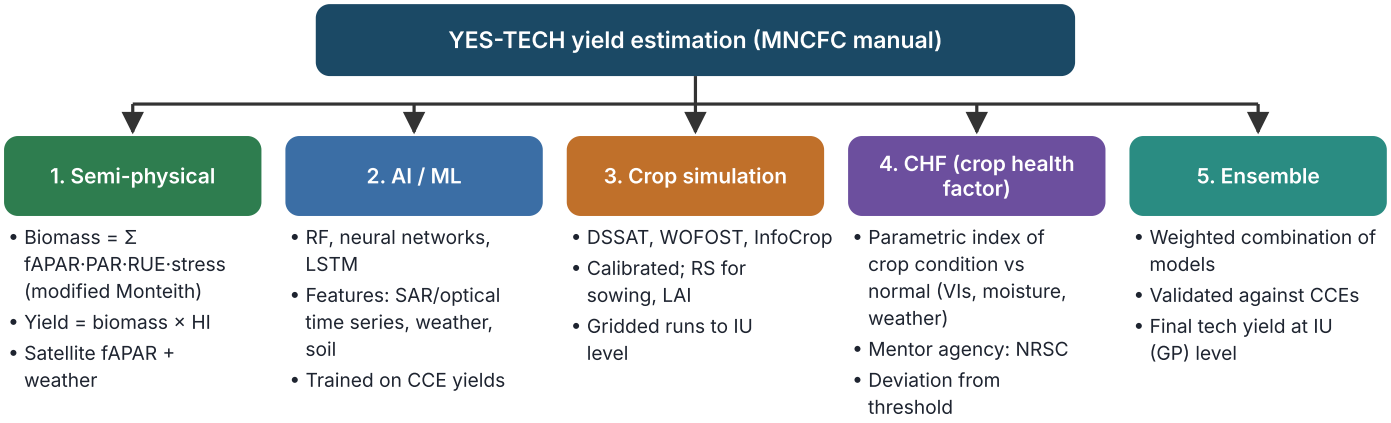
- Weather index: payout when rainfall, temperature, humidity cross triggers
- Your notes: weather-based for horticultural crops
- Basis risk: index ≠ farm loss
- Satellite rainfall and LST improve trigger design

Interview one-liner: PMFBY pays when the IU's actual yield falls below threshold; YES-TECH estimates that yield from satellite-driven models combined in an ensemble and blends it 30:70 with CCE yield, cutting delays and disputes.

Key papers / sources: PMFBY Operational Guidelines (MoA&FW, 2016, revised 2020) · YES-TECH Manual (MoA&FW, 2023) · Lok Sabha reply on 30% YES-TECH weightage · Dutta et al., MNCFC smart sampling studies

PMFBY payout and YES-TECH models in detail

Your notes ask how a payout is calculated, where CCEs and smart sampling are used, how tech yield and CCE yield are weighted, and what the crop health factor (CHF) is. YES-TECH's manual (MNCFC) recommends several model families combined in an ensemble.



Step	Worked example (illustrative numbers)
Threshold yield (TY)	Average of best 5 of last 7 years = 5.0 t/ha × indemnity level 80% = **4.0 t/ha**
Actual yield (AY), blended	0.7 × CCE yield 3.2 + 0.3 × tech yield 3.6 = **3.32 t/ha**
Shortfall	(TY − AY)/TY = (4.0 − 3.32)/4.0 = **17%**
Claim	17% × sum insured ₹60,000/ha = **₹10,200 per ha**, paid to all insured farmers of the IU for that crop
CHF route (per state presentations)	70% weight to CCE-yield deviation from TY + 30% weight to CHF deviation from TY

Where CCEs and smart sampling fit

- CCEs (crop-cutting experiments) measure yield in sample plots of each IU
- Insurance unit: village/gram panchayat for major crops
- Smart sampling: satellite crop-vigour strata decide where CCEs go, so fewer plots represent the IU better
- Food crops: area-yield index; horticulture: weather-based (RWBCIS)

Designing products on other factors

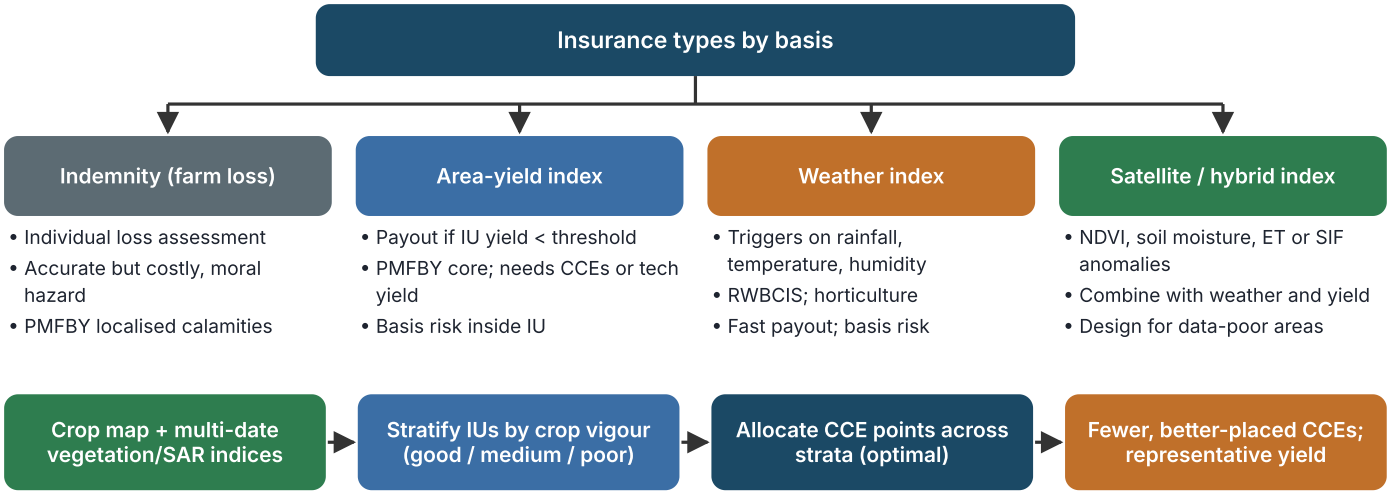
- Weather index: rainfall deficit/excess, temperature, humidity triggers
- Satellite index: NDVI, VCI, soil moisture deviation (as in drought declaration)
- Hybrid: weather + satellite + yield
- Goal: low basis risk, quick payout, fewer disputes

Interview one-liner: YES-TECH estimates insurance-unit yield with semi-physical, AI/ML, crop simulation and crop-health-factor approaches combined in an ensemble; the tech yield is blended 30:70 with CCE yield, and the claim is the percentage shortfall below threshold yield times the sum insured.

Key papers / sources: YES-TECH Manual for Implementation (MNCFC, DA&FW, 2023) · APSAC YES-TECH presentation (CHF weighting) · PMFBY Operational Guidelines (2020) · Monteith (1977) Phil. Trans. R. Soc. B 281:277 (RUE)

Remote sensing in crop insurance: smart sampling and insurance design

Your notes list two main uses of remote sensing in insurance — smart sampling of CCEs and yield estimation — plus designing insurance for new areas. Insurance types differ by the 'basis' used to trigger payouts.



Smart sampling of crop-cutting experiments

Designing insurance (your 'we can do')

- Delineate homogeneous risk zones from long-term satellite NDVI/rainfall/yield
- Set IU size so variance within IU is small (less basis risk)
- Estimate historic loss frequency → premium rating
- Choose index: yield, weather or satellite by data availability
- Validate triggers on past droughts/floods

Other RS uses

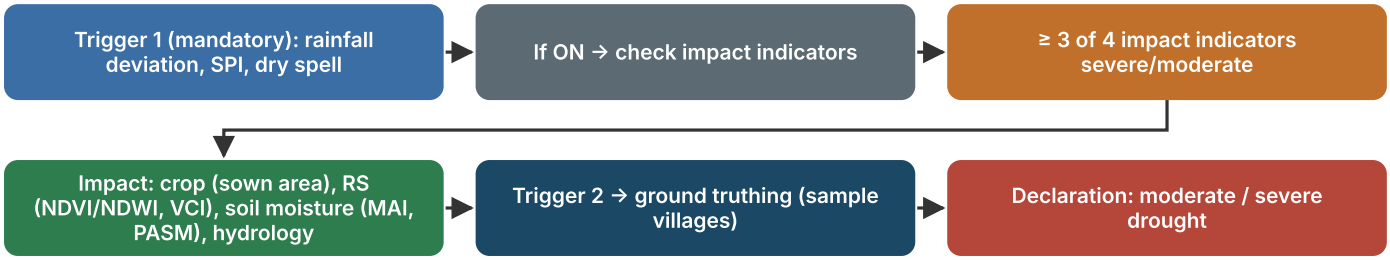
- Sowing and area discrepancy checks (proposal vs crop map)
- Prevented sowing assessment
- Mid-season adversity (drought, flood, hail) using SAR/optical change
- Post-harvest losses (unseasonal rain) mapping
- Claims dispute resolution with tech yield

Interview one-liner: Remote sensing serves insurance by stratifying insurance units for smart CCE sampling, estimating yield directly (YES-TECH), detecting mid-season adversity, and designing index products with low basis risk from long-term satellite records.

Key papers / sources: Carter et al. (2017) Annu. Rev. Resour. Econ. 9:421 (index insurance) · Ray et al. (2014) ISPRS Archives (RS in Indian crop insurance) · Skees (2008) Agric. Finance Review 68:197 · PMFBY / YES-TECH guidelines

Agricultural drought monitoring (Manual for Drought Management, 2016)

India's drought declaration follows the Manual for Drought Management (DAC&FW, December 2016). Rainfall is the mandatory first trigger; drought intensity is then judged from impact indicators including remote sensing, and confirmed by field verification. MNCFC's NADAMS supplies satellite-based assessments.



Drought declaration logic (Manual 2016)

Indicator	Formula / meaning
SPI	Rainfall standardised against gamma-fitted long-term distribution (McKee et al. 1993)
NDVI / NDWI (LSWI)	Greenness and canopy water; deviation from normal
VCI	$(NDVI - NDVI_{min})/(NDVI_{max} - NDVI_{min}) \times 100$ (Kogan 1995); < 40 indicates stress
TCI / VHI	Thermal condition from LST; VHI = 0.5·VCI + 0.5·TCI
MAI	AET/PET (moisture adequacy) from water balance
PASM	Percent available soil moisture (model or satellite)
SIF, ET anomalies	Newer: detect stress earlier than NDVI

Types of drought

- Meteorological: rainfall deficit
- Agricultural: soil moisture/crop stress
- Hydrological: reservoirs, groundwater, streamflow
- Socio-economic: supply–demand impact

RS products

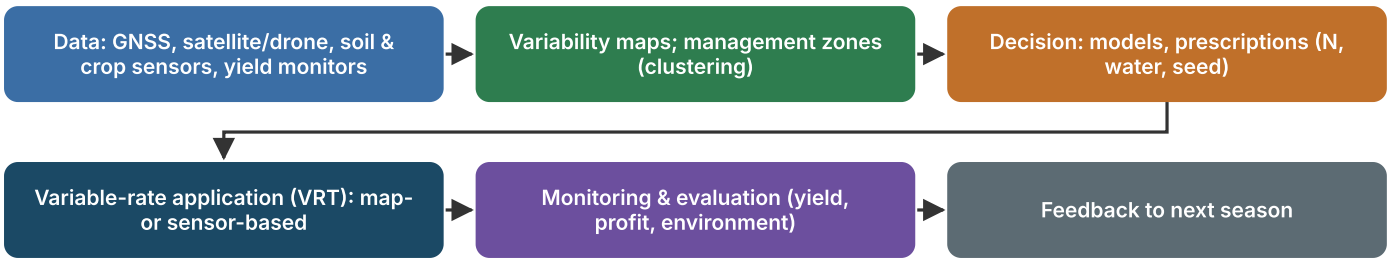
- NADAMS (since 1989, now MNCFC): district/sub-district monthly reports
- MODIS/AWiFS/Sentinel-2 NDVI, LSWI; SMAP/NISAR soil moisture
- Kalpana/INSAT LST, MOSDAC products
- Studies: framework misses some regional droughts; SIF can help

Interview one-liner: Drought is declared when the rainfall trigger is on and at least three of four impact indicators — crop, remote sensing (NDVI/NDWI/VCI), soil moisture (MAI/PASM) and hydrology — show moderate or severe conditions, confirmed by ground truthing.

Key papers / sources: Manual for Drought Management (DAC&FW, 2016) · McKee, Doesken & Kleist (1993) 8th Conf. Appl. Climatol. · Kogan (1995) Adv. Space Res. 15:91 · Gao (1996) RSE 58:257 · Mishra & Singh (2010) J. Hydrol. 391:202 (drought review)

Precision agriculture: principles, VRT and Indian context

Precision agriculture manages within-field variability: the right input, at the right place, time and rate. Your notes highlight SAC's precision agriculture work, principles applicable in India, modelling tools and variable-rate technology (VRT); ICAR runs the Network Programme on Precision Agriculture (NePPA).



Precision agriculture cycle

Principles

- Spatial and temporal variability exists and matters
- Measure it (sensing), understand it (analysis), act on it (VRT)
- Management zones: areas with similar yield-limiting factors
- Economic and environmental optimisation

Tools

- NDVI sensors (GreenSeeker) and leaf colour chart for N
- Drones for spraying and imaging; Sentinel-2 for zones
- Soil EC/pH mapping; yield monitors
- Crop models and decision support (DSS)

Indian context

- Small holdings (≈1.08 ha average) → community/custom hiring, drones
- ICAR-NePPA launched Sept 2021, IARI lead, 16 institutes
- Namo Drone Didi, Digital Agriculture Mission
- SAC: precision agriculture studies using hyperspectral and high-resolution data

VRT type	How it works	Example
Map-based	Prescription map made before application (soil tests, previous yield, satellite)	Variable fertiliser by zone
Sensor-based (real time)	On-the-go sensor reads crop and adjusts rate	Optical N sensor on spreader
Hybrid	Map + real-time correction	Drone imagery + tractor sensor

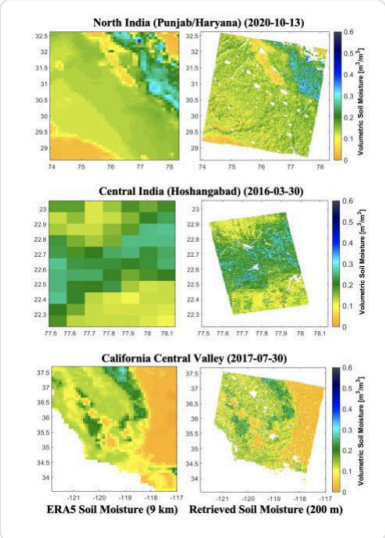
Interview one-liner: Precision agriculture measures within-field variability with satellite, drone and proximal sensors, defines management zones and applies inputs at variable rates; in India it is adapted to small farms through drones, custom hiring and ICAR's NePPA programme.

Key papers / sources: Zhang, Wang & Wang (2002) Comput. Electron. Agric. 36:113 · Gebbers & Adamchuk (2010) Science 327:828 · Mulla (2013) Biosyst. Eng. 114:358 · ICAR-NePPA launch note (2021)

Current satellite missions to know

Interviewers often ask which current missions matter for agriculture. Know the band, resolution, revisit and the agricultural products of each.

Mission	Key facts	Agricultural use
NISAR (NASA–ISRO)	Launched 30 Jul 2025 (GSLV-F16); L-band (NASA, 24 cm) + S-band (ISRO, ~10 cm) SAR; 12-day repeat; ~240 km swath; 3–10 m	Crop area and type (all-weather), soil moisture (L3 product ~200 m), biomass, crop growth, flood
FLEX (ESA EE-8)	Launched 15 Sep 2026 with Sentinel-3C; FLORIS spectrometer ~500–780 nm, 300 m; tandem with Sentinel-3	Sun-induced fluorescence → photosynthesis, early stress
EOS-04 (RISAT-1A)	C-band SAR, launched Feb 2022	Kharif crop mapping (FASAL), flood, soil moisture
Resourcesat-2/2A	LISS-IV 5.8 m, LISS-III 23.5 m, AWiFS 56 m	Crop area, CHAMAN orchards, crop condition
INSAT-3D/3DR/3DS	Geostationary imager + sounder; 4 km split-window TIR; 3DS launched Feb 2024	Rainfall, LST, insolation, fog, cyclone; ET and drought
EOS-06 (Oceansat-3), HySIS	Ocean colour/scatterometer; HySIS hyperspectral (2018)	Weather inputs; crop/soil hyperspectral studies
TRISHNA (ISRO–CNES)	Planned thermal mission, ~57 m TIR, few-day revisit	Field-scale ET, crop water stress, irrigation
Sentinel-1/2, Landsat-8/9	C-band SAR 12-day; 10–20 m optical with red-edge (5-day); 30 m + 100 m TIR	Workhorse data for YES-TECH, FASAL, residue burning, ET
SMAP, SMOS, ECOSTRESS, GEDI	L-band radiometer SM; ISS thermal 70 m; space lidar canopy height	Soil moisture, ET, biomass/carbon



NISAR soil moisture algorithm tests over Punjab–Haryana and Central India (from NISAR L3 SM ATBD; earlier deck)

Interview one-liner: NISAR, launched in July 2025, is the first dual-frequency (L- and S-band) SAR with a 12-day repeat, giving all-weather crop mapping and 200 m soil moisture; FLEX, launched in September 2026, maps sun-induced fluorescence to track photosynthesis directly.

Key papers / sources: NISAR Science Users' Handbook (JPL, 2018) · NISAR L3 Soil Moisture ATBD (JPL D-107679) · Drusch et al. (2017) IEEE TGRS 55:1273 · ISRO mission pages (EOS-04, INSAT-3DS)

SAR frequency bands: how NISAR's L and S bands differ

Radar wavelength controls what the signal 'sees'. Longer waves penetrate deeper into canopies and soil and interact with larger structures (stems, branches); shorter waves scatter from leaves and the canopy top. NISAR is the first mission to fly both L-band (NASA) and S-band (ISRO) SAR together.

Band	Wavelength / frequency	Interacts with	Example missions
X	2.4–3.75 cm (8–12.5 GHz)	Leaves, canopy top, surface texture; very high resolution	TerraSAR-X, COSMO-SkyMed
C	3.75–7.5 cm (4–8 GHz); typical 5.6 cm	Upper canopy, crop structure; saturates at moderate biomass	Sentinel-1, EOS-04 (RISAT-1A), RADARSAT
S	7.5–15 cm (2–4 GHz); NISAR ≈ 9.4 cm (3.2 GHz)	Between C and L: crops and shrubs, moderate biomass	NISAR (ISRO S-SAR), NovaSAR
L	15–30 cm (1–2 GHz); NISAR ≈ 24 cm (1.26 GHz)	Penetrates canopy to stems, branches and soil; soil moisture under vegetation; forest biomass; deformation	NISAR (NASA L-SAR), ALOS-2 PALSAR-2, SAOCOM
P	30–100 cm (~70 cm, 435 MHz)	Trunks; dense forest biomass	ESA BIOMASS (launched 2025)



Penetration and sensitivity increase with wavelength

NISAR specifics

- Launched 30 July 2025 on GSLV-F16
- SweepSAR: ~240 km swath at 3–10 m resolution
- 12-day repeat (6-day with ascending + descending)
- Dual/quad polarisation; free, open data

Why dual frequency helps agriculture

- S-band sensitive to crop growth stages and biomass of crops
- L-band gives soil moisture under vegetation and woody biomass
- Ratio of L and S responses separates crop types and structure
- Longer coherence at L-band for interferometry (subsidence, deformation)

Trade-offs

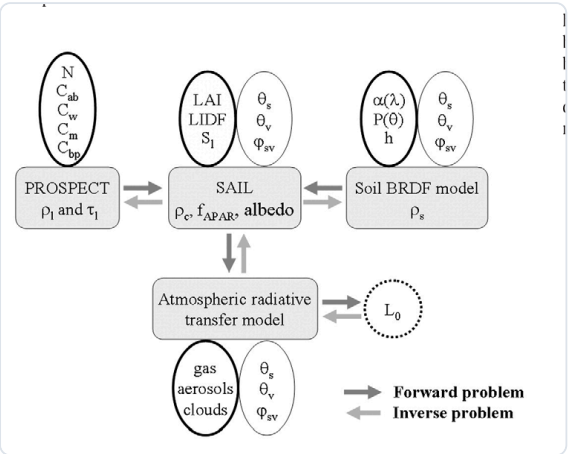
- L-band affected by ionosphere (Faraday rotation)
- Lower resolution at longer λ for same antenna
- C-band better for small crop changes, frequent revisit (Sentinel-1)
- Speckle in all SAR: multilooking, filters

Interview one-liner: Penetration and the size of features a radar senses scale with wavelength: C-band stays near the canopy top, S-band (NISAR-ISRO, ~9 cm) suits crops and moderate biomass, and L-band (NISAR-NASA, ~24 cm) reaches stems and soil, giving soil moisture under vegetation and forest biomass.

Key papers / sources: NISAR Science Users' Handbook (JPL, 2018) · Ulaby, Moore & Fung (1986) Microwave Remote Sensing, Vols I–III · Woodhouse (2006) Introduction to Microwave Remote Sensing · McNairn & Brisco (2004) Can. J. Remote Sens. 30:525 (SAR for agriculture)

Crop parameter retrieval: phenology, LAI, chlorophyll, N, leaf water

Biophysical (LAI, fAPAR, canopy water) and biochemical (chlorophyll, N) parameters are retrieved by empirical indices, physical radiative transfer models (RTM) such as PROSAIL, or hybrid machine learning trained on RTM simulations.



PROSPECT (leaf) + SAIL (canopy) = PROSAIL; forward and inverse problems (Jacquemoud et al. 2009)

Parameter	Typical indices / method	Sensor bands
Phenology (SOS, peak, EOS)	Smoothed NDVI/EVI time series (Savitzky–Golay, TIMESAT); thresholds or derivative	MODIS, Sentinel-2, Sentinel-1 (VH backscatter)
LAI, fAPAR	NDVI/EVI regressions; PROSAIL inversion (LUT, ANN, GA); hybrid GPR	Red, NIR, red-edge
Chlorophyll / CCC	Red-edge CI = NIR/RE − 1 (Gitelson 2003); MTCI; TCARI/OSAVI	Sentinel-2 B5–B7, hyperspectral
Nitrogen	Via CCC and red-edge position; N nutrition index	Red-edge, hyperspectral (PRISMA, EnMAP, HySIS)
Leaf / canopy water	NDWI/LSWI = (NIR − SWIR)/(NIR + SWIR) (Gao 1996); EWT from RTM	SWIR 1.6, 2.2 µm
Crop type & area	Multi-temporal classification (RF, SVM, CNN); SAR backscatter profiles	Sentinel-1/2, EOS-04

Retrieval families (Verrelst 2015)

- **Parametric:** VI regressions — simple, site-specific
- **Non-parametric:** ML (RF, ANN, GPR) — accurate, need data
- **Physical:** RTM inversion — generic, ill-posed (LAI–leaf angle compensation)
- **Hybrid:** ML trained on RTM simulations — operational LAI products

IARI wheat studies (earlier deck)

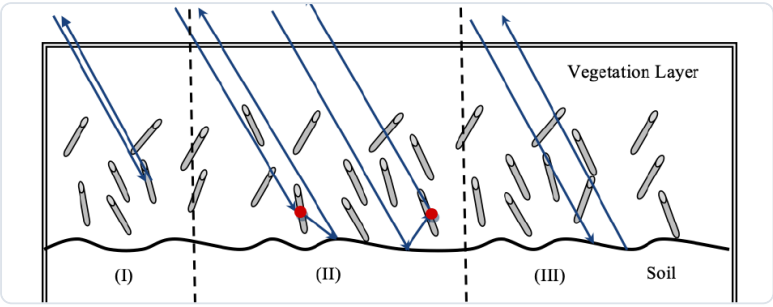
- PROSAIL validated for wheat BRF (Chakraborty 2015)
- GA inversion: LAI $R^2 \approx 0.9$ (Mridha 2014)
- LUT vs ANN: LAI RMSE ≈ 0.5; LISS-III LAI map, Sheopur (Sehgal 2016)
- Accuracy order: LAI > CCC > Cab > Cw

Interview one-liner: Crop parameters are retrieved with vegetation indices, radiative transfer model inversion (PROSAIL by LUT, ANN or GA), or hybrid ML trained on RTM simulations; red-edge bands carry chlorophyll and N, SWIR carries leaf water.

Key papers / sources: Jacquemoud et al. (2009) RSE 113:S56 · Verrelst et al. (2015) ISPRS J. 108:273 · Gitelson et al. (2003) J. Plant Physiol. 160:271 · Dash & Curran (2004) IJRS 25:5403 · Jönsson & Eklundh (2004) Comput. Geosci. 30:833 · Sehgal et al. (2016) Inf. Process. Agric. 3:107

Soil moisture retrieval with microwave remote sensing

Microwaves respond to soil moisture because water's dielectric constant (~80) is far higher than dry soil's (~3–5). Passive radiometers (SMAP, SMOS) measure brightness temperature at coarse resolution; active SAR (NISAR, Sentinel-1, EOS-04) measures backscatter at fine resolution but is sensitive to roughness and vegetation.



Backscatter contributions: (I) canopy volume, (II) canopy–soil interaction, (III) soil surface

SAR σ° (VV, VH) or radiometer TB → Remove vegetation: water cloud model / τ-ω → Soil σ° → dielectric ε (IEM, Oh, Dubois, change detection) → ε → θ (Topp / Dobson mixing) → Soil moisture map; validate with in-situ

Key equations

- Water cloud model: $\sigma^\circ_{total} = A \cdot V_1 \cdot \cos\theta(1 - \tau^2) + \tau^2 \cdot \sigma^\circ_{soil}$, $\tau^2 = \exp(-2B \cdot V_2/\cos\theta)$
- Passive τ-ω: $T_B = T_s e_s \gamma + T_c(1 - \omega)(1 - \gamma)(1 + R_s\gamma)$
- Change detection: $\theta \propto (\sigma^\circ - \sigma^\circ_{dry})/(\sigma^\circ_{wet} - \sigma^\circ_{dry})$
- Topp: $\theta = -5.3\times10^{-2} + 2.92\times10^{-2}\varepsilon - 5.5\times10^{-4}\varepsilon^2 + 4.3\times10^{-6}\varepsilon^3$

Band choice and NISAR

- L-band (24 cm) penetrates vegetation better than C-band; senses ~5 cm
- NISAR L3 SM: time-series ratio, multiscale fusion; ~200 m, 12-day
- Sentinel-1 + Sentinel-2 (NDVI for vegetation term) at field scale
- Uses: drought PASM, irrigation, yield models, insurance

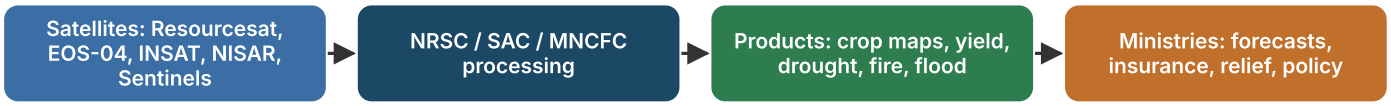
Interview one-liner: Soil moisture is retrieved from microwave signals because water raises the soil dielectric constant; SAR backscatter is corrected for vegetation (water cloud model) and roughness (IEM, change detection), then converted to moisture with dielectric mixing models, and NISAR's L-band now gives 200 m products.

Key papers / sources: Karthikeyan et al. (2017) Adv. Water Resour. 109:106 & 236 · Attema & Ulaby (1978) Radio Sci. 13:357 · Wagner et al. (1999) RSE 70:191 · Entekhabi et al. (2010) Proc. IEEE 98:704 (SMAP) · Kerr et al. (2010) Proc. IEEE 98:666 (SMOS) · Topp et al. (1980) WRR 16:574

Remote-sensing-based government programmes: crops, water and disasters

Satellite data are built into India's main agricultural programmes, from crop forecasting and insurance to drought and flood assessment.

Programme	Agency	Remote sensing role
FASAL	MNCFC (SAC methods)	Multiple in-season area, yield, production forecasts (optical + SAR)
CHAMAN	MNCFC, SAC, NRSC, NESAC	Horticulture area, production, site suitability
NADAMS	MNCFC (since 1989 at NRSC)	Agricultural drought assessment (NDVI, LSWI, VCI, soil moisture)
PMFBY: YES-TECH, CROPIC, WINDS	MoA&FW, MNCFC	Tech-based yield (30% weight), smart sampling, crop photos, weather network
KISAN pilot (2015)	MoA&FW with ISRO	Crop insurance using space technology (forerunner of YES-TECH)
Crop Residue Management / CREAMS	MoA&FW; ICAR-IARI	Daily satellite fire bulletins, burnt area
Flood and cyclone damage	NRSC Decision Support Centre	SAR flood maps, crop damage for relief and insurance
Agromet Advisory (GKMS)	IMD with ICAR/SAUs	Satellite rainfall, soil moisture, crop condition in advisories
India-WRIS, Jal Shakti water body mapping	CWC, NRSC	Reservoir and water body monitoring
SVAMITVA	MoPR, Survey of India	Drone survey of village land parcels (property cards)



From satellites to policy

Interview tips

- Know the acronyms and the agency for each
- Link each to the sensor type (optical vs SAR vs thermal)
- Mention accuracy needs: area ±5%, yield at IU level
- Mention new missions that improve each programme

Recent additions

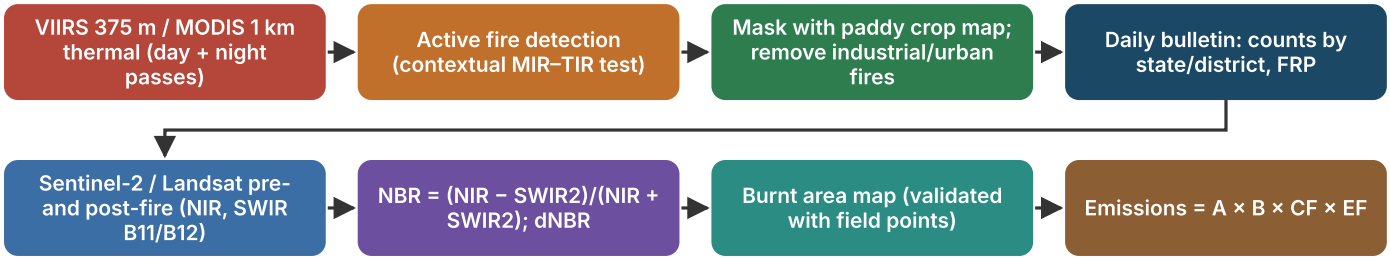
- Digital Agriculture Mission: AgriStack, Krishi-DSS, digital crop survey
- Namo Drone Didi: drones for spraying
- NISAR soil moisture and crop products
- Mission Mausam (2024): better forecasts for advisories

Interview one-liner: FASAL, CHAMAN and NADAMS at MNCFC, YES-TECH under PMFBY, CREAMS fire monitoring and NRSC's flood damage mapping are the main satellite-based programmes for crops and disasters.

Key papers / sources: MNCFC programme pages · PMFBY & YES-TECH guidelines · NRSC Decision Support Centre · IMD Agromet Advisory Service documents · PIB releases

Mapping crop residue burning: active fires, FRP and burnt area

After the kharif paddy harvest (Oct–Nov) farmers in Punjab, Haryana, UP and neighbouring states burn straw to clear fields for wheat. IARI's CREAMS laboratory issues daily bulletins of active fire events from VIIRS and MODIS received at the IARI satellite ground station; burnt area is mapped from Sentinel-2 SWIR bands.



Two complementary approaches: (1) active fire counts and intensity, (2) burnt area

Quantity	Meaning	Note
Active fire	Pixel with fire burning at overpass (MIR 4 µm brightness jump)	Misses fires between overpasses or under smoke/cloud; small fires may be missed by 1 km MODIS
FRP (MW)	Fire radiative power ∝ $\sigma(T_f^4 - T_b^4)$ of the fire	Integrated over time → FRE (MJ); biomass burned ≈ 0.368 kg MJ^{-1} (Wooster 2005)
NBR / dNBR	Burn lowers NIR and raises SWIR	dNBR = $NBR_{pre} - NBR_{post}$; severity classes
Emission	A × fuel load B × combustion factor CF × emission factor EF	Seiler & Crutzen (1980) approach; PM2.5, CO, CO_2

CREAMS bulletin (IARI)

- Coverage: Punjab, Haryana, UP, Rajasthan, MP, Delhi
- VIIRS (Suomi-NPP, NOAA-20) + MODIS (Terra, Aqua)
- District/block counts, coordinates, time, fire power
- Standard protocol for satellite fire monitoring (2021)

Why vegetation indices matter

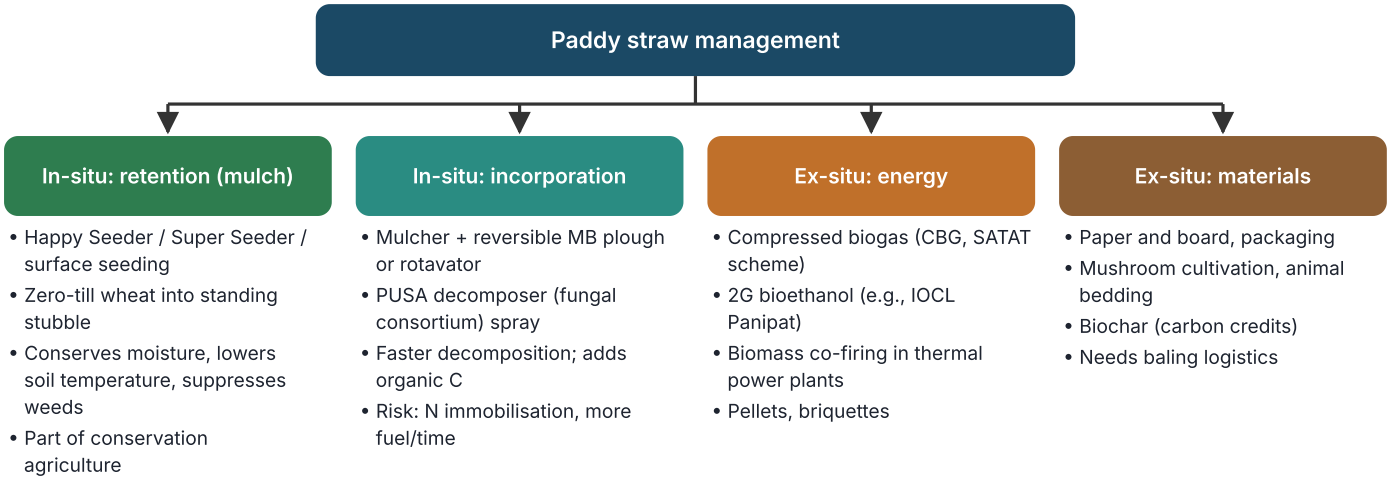
- Paddy crop map separates residue fires from other fires
- Harvest detection from NDVI drop
- Post-burn: char lowers NIR; NBR more sensitive than NDVI
- Smoke trapped in shallow winter boundary layer → Delhi air quality

Interview one-liner: Residue burning is monitored as daily active-fire counts and fire radiative power from VIIRS and MODIS thermal bands (CREAMS, IARI), and as burnt area from the post-burn drop in Sentinel-2 NBR; FRP integrated over time estimates the biomass burned.

Key papers / sources: Giglio et al. (2016) RSE 178:31 (MODIS C6 fires) · Schroeder et al. (2014) RSE 143:85 (VIIRS 375 m) · Wooster et al. (2005) JGR 110:D24311 (FRP) · Key & Benson (2006) FIREMON (NBR) · Seiler & Crutzen (1980) Climatic Change 2:207 · Vadrevu et al. (2011) Environ. Pollut. 159:1560 · CREAMS-IARI bulletins

Crop residue management: in-situ vs ex-situ, incorporation vs retention

Alternatives to burning are either in-situ (straw stays in the field, incorporated or retained as mulch) or ex-situ (straw is baled and used elsewhere). IARI's PUSA decomposer speeds up in-field decomposition.



Aspect	Residue incorporation	Residue retention (surface)
Operation	Chop and mix into soil (tillage)	Leave on surface; seed with Happy Seeder
Decomposition	Faster (soil contact, PUSA decomposer)	Slower; acts as mulch
Soil effects	Raises SOC in plough layer; temporary N tie-up	Moisture conservation, erosion control, cooler soil, weed suppression
Cost/time	More tractor passes and fuel; delays sowing	One pass; fits short window before wheat
Issues	N immobilisation; pest carry-over	Machine availability; rodents; uneven germination if heavy load

Remote sensing links

- Fire counts and burnt area track adoption of alternatives
- Residue cover indices (CAI, NDTI) separate mulched vs burnt vs tilled fields
- Zero-till detection → conservation agriculture mapping (next page)

Policy and programmes

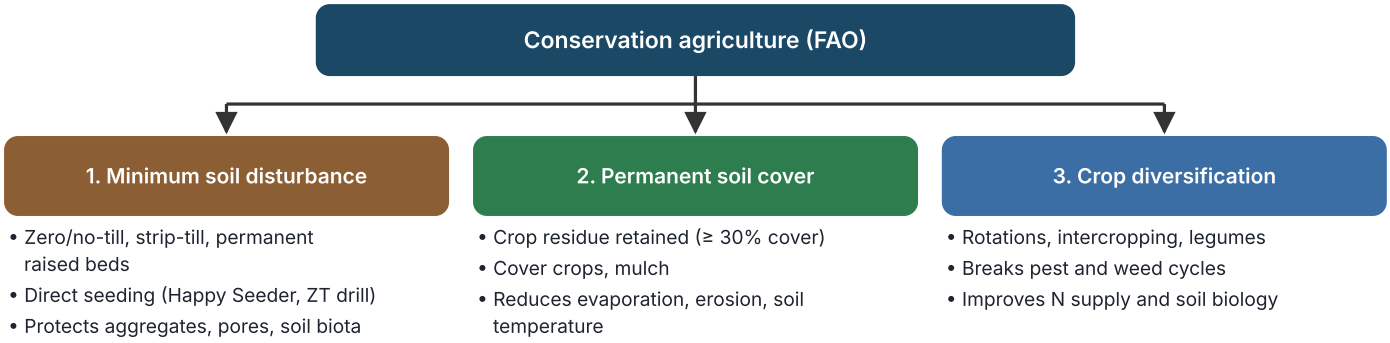
- Crop Residue Management (CRM) scheme (2018): machinery subsidy, custom hiring
- CAQM directions for NCR; ex-situ supply chains
- Engineering work at IARI on residue machinery (your notes)

Interview one-liner: In-situ options keep straw in the field — retained as mulch with a Happy Seeder or incorporated with the PUSA decomposer — while ex-situ options use it for compressed biogas, 2G ethanol, power and biochar; retention conserves moisture, incorporation decomposes faster but can tie up nitrogen.

Key papers / sources: Sidhu et al. (2007) Field Crops Res. 103:1 (Happy Seeder) · Jat et al. (2020) Nature Sustainability 3:336 (residue & CA, IGP) · Bhuvaneshwari, Hettiarachchi & Meegoda (2019) IJERPH 16:832 · ICAR-IARI PUSA decomposer technical bulletins

Conservation agriculture: principles, practices and trade-offs

FAO defines conservation agriculture (CA) as a farming system built on three linked principles applied together: minimum mechanical soil disturbance, permanent organic soil cover, and diversified crop rotations. In the rice–wheat system of the Indo-Gangetic Plains, CA mainly means zero-till wheat after rice with residue retention.



Aspect	Conventional tillage	Conservation agriculture
Tillage	Repeated ploughing, puddling	Zero/minimum tillage
Residue	Removed or burnt	Retained on surface
Wheat sowing after rice	Delayed by field preparation	Timely (direct drilling into stubble)
Soil	Lower SOC, crusting, compaction pan	Higher surface SOC, aggregation, infiltration, biota
Water and energy	More irrigation, fuel and labour	Saves water, diesel, labour, cost
Emissions	Burning, more CO_2 from tillage and fuel	No burning; possible soil C gain

Related practices

- Direct-seeded rice (DSR) instead of puddled transplanting
- Laser land levelling (water saving)
- Permanent beds and furrow irrigation
- Brown manuring, legume inclusion
- Precision nutrient management (LCC, GreenSeeker)

Challenges

- Weed shift and herbicide dependence
- Residue handling; machine availability (Happy Seeder)
- Initial N immobilisation; possible early yield penalty
- Rodents, pests in residue; knowledge gap
- Pittelkow et al. (2015): no-till alone lowers yield; full CA reduces the gap, best in dry areas

Facts to quote

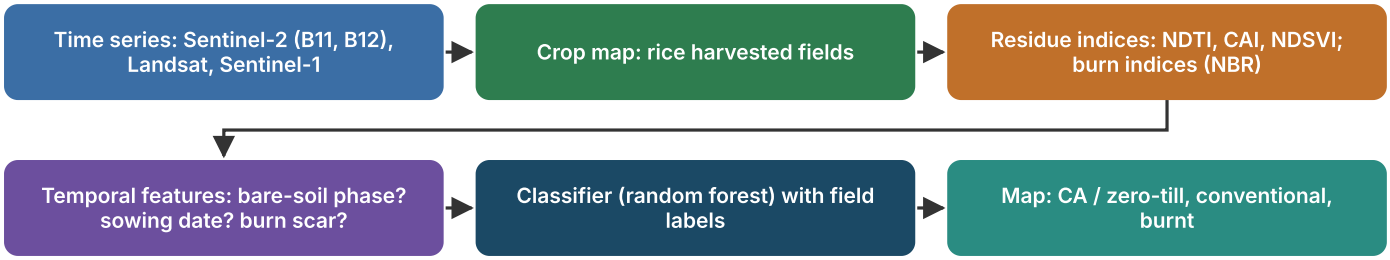
- Global CA ≈ 180 Mha (2015–16; Kassam et al. 2019)
- IGP rice–wheat: CIMMYT/ICAR Rice–Wheat Consortium spread zero-till wheat
- CA links to residue-burning reduction and carbon credits
- Remote sensing can map CA adoption (next page)

Interview one-liner: Conservation agriculture combines minimum soil disturbance, permanent residue cover and crop diversification; in the rice–wheat system it means zero-till wheat drilled into retained paddy straw, which saves water, fuel and time and avoids burning, though weeds, residue handling and early yield penalties are challenges.

Key papers / sources: Hobbs, Sayre & Gupta (2008) Phil. Trans. R. Soc. B 363:543 · Kassam et al. (2019) Int. J. Environ. Stud. 76:29 · Pittelkow et al. (2015) Nature 517:365 · Jat et al. (2020) Nature Sustainability 3:336 · FAO Conservation Agriculture pages

Identifying conservation agriculture fields from satellite data

Conservation agriculture (CA) rests on three FAO principles: minimum soil disturbance (no-till), permanent soil cover (≥30% residue) and crop diversification. Satellites can detect residue cover and the absence of a tillage/bare-soil phase. Your notes refer to a paper by Dr. Rajkumar on this topic — add its details.



General workflow for mapping tillage and residue practice

Index	Formula	Basis
CAI (cellulose absorption index)	$100 \times [0.5(R_{2.0} + R_{2.2}) - R_{2.1}]$	Cellulose–lignin absorption at 2.1 µm (hyperspectral)
NDTI (normalised difference tillage index)	(SWIR1 − SWIR2)/(SWIR1 + SWIR2)	Residue raises SWIR1 relative to SWIR2; works with Sentinel-2 B11/B12, Landsat
NDSVI	(SWIR1 − Red)/(SWIR1 + Red)	Senesced vegetation / residue
SINDRI	(ASTER 6 − 7)/(6 + 7)	Lignin–cellulose (Serbin 2009)

Detecting zero-till wheat

- No ploughing → no bare, dark soil phase between rice harvest and wheat
- Earlier wheat sowing under Happy Seeder
- Residue signal persists (high NDTI) through crown root stage
- No burn scar (NBR unchanged)

Challenges

- Residue moisture lowers CAI/NDTI contrast
- Small fields, mixed pixels; cloud at sowing
- Need ground truth of tillage practice
- Zhou et al. (2021): ~75% accuracy for tillage maps in IGP with Sentinel-2 + RF

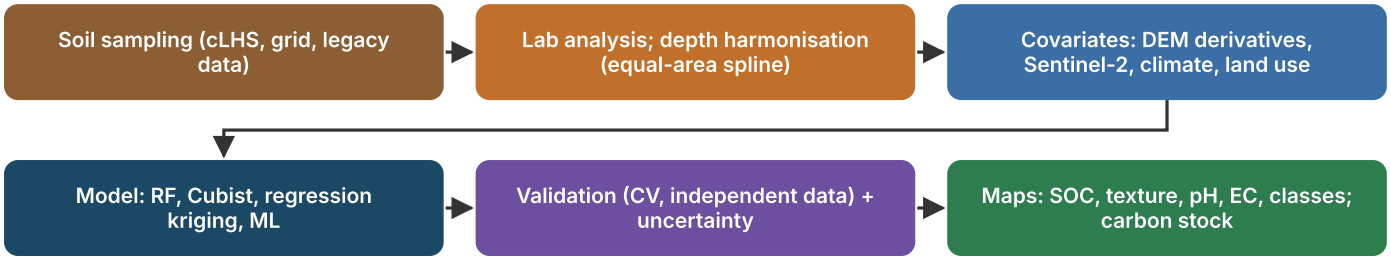
Interview one-liner: CA fields are identified by persistent residue cover (high NDTI/CAI from SWIR bands), the absence of a bare-soil tillage phase and burn scars, and early sowing, combined in a time-series classifier such as random forest.

Key papers / sources: Daughtry et al. (2005) Agron. J. 97:864 (CAI) · van Deventer et al. (1997) PE&RS 63:87 (NDTI) · Zheng et al. (2014) Soil Tillage Res. 138:26 (review) · Zhou et al. (2021) Remote Sens. 13:5108 (IGP tillage mapping) · ICAR-IARI ISPRS Archives XLII-3/W6:423 (2019) residue cover · FAO CA principles

Digital soil mapping

Digital soil mapping (DSM) predicts soil properties or classes at unsampled locations from field observations and environmental covariates, using the scorpan model. It replaces hand-drawn polygon maps with quantitative maps that come with uncertainty. Your notes mention Dr. Bappa Das's DSM work.

S = f(s, c, o, r, p, a, n) + ε (soil, climate, organisms, relief, parent material, age, space)



DSM workflow (Minasny & McBratney)

Methods

- Regression kriging: trend from covariates + kriged residual
- Machine learning: random forest, Cubist, gradient boosting
- Geostatistics: ordinary/universal kriging, co-kriging
- GlobalSoilMap specs: 6 depths (0–5, 5–15, 15–30, 30–60, 60–100, 100–200 cm), 90 m

Covariates from remote sensing

- DEM: slope, TWI, curvature (relief)
- Bare-soil composites (Sentinel-2, Landsat) for texture, SOC
- Hyperspectral and vis-NIR spectroscopy
- Climate: rainfall, temperature; land use from classification

Uses

- Soil carbon stocks and change (MRV for carbon credits)
- Salinity/sodicity mapping
- Land capability, fertiliser recommendations
- Inputs for crop models and digital twins

Interview one-liner: Digital soil mapping predicts soil properties from field samples and environmental covariates through the scorpan model, using machine learning or regression kriging, and reports uncertainty with each map.

Key papers / sources: McBratney, Mendonça Santos & Minasny (2003) Geoderma 117:3 · Minasny & McBratney (2016) Geoderma 264:301 · Minasny et al. (2013) Adv. Agron. 118:1 · Arrouays et al. (2014) Adv. Agron. 125:93 (GlobalSoilMap)

Remote sensing of soil salinity and alkalinity (sodicity)

India has about 6.7 million ha of salt-affected soils, roughly 3 Mha saline and 3.8 Mha sodic (CSSRI estimates). Saline soils show white salt crusts and stressed crops; sodic (alkali) soils have high pH and exchangeable sodium, poor structure and often a puffy or dark surface.

Class	EC_e (dS m^{-1})	ESP	pH	Surface feature
Saline	> 4	< 15	< 8.5	White salt crust, high visible reflectance
Sodic (alkaline)	< 4	> 15	> 8.5	Dispersed, crusted; dark (organic) or whitish; poor drainage
Saline-sodic	> 4	> 15	variable	Mixed features



Index	Formula	Sensitive to
SI	√(Blue × Red)	Salt crust brightness
SI1, SI2	√(Green × Red); √($Green^2$ + Red^2 + NIR^2)	Bright saline surfaces
NDSI	(Red − NIR)/(Red + NIR)	Salinity (inverse of NDVI)
BI	√(Red^2 + NIR^2)	Bare, bright soils
NDVI, EVI, SAVI	Vegetation vigour	Indirect: crop stress on saline/sodic land
SWIR bands	1.4, 1.9, 2.2 µm features	Gypsum, clay minerals, sodicity

Approaches

- Direct: bare soil reflectance and indices
- Indirect: vegetation stress (halophytes, poor crop)
- Microwave: salt raises imaginary dielectric constant
- Multi-temporal: dry-season (pre-monsoon) imagery best

Limitations

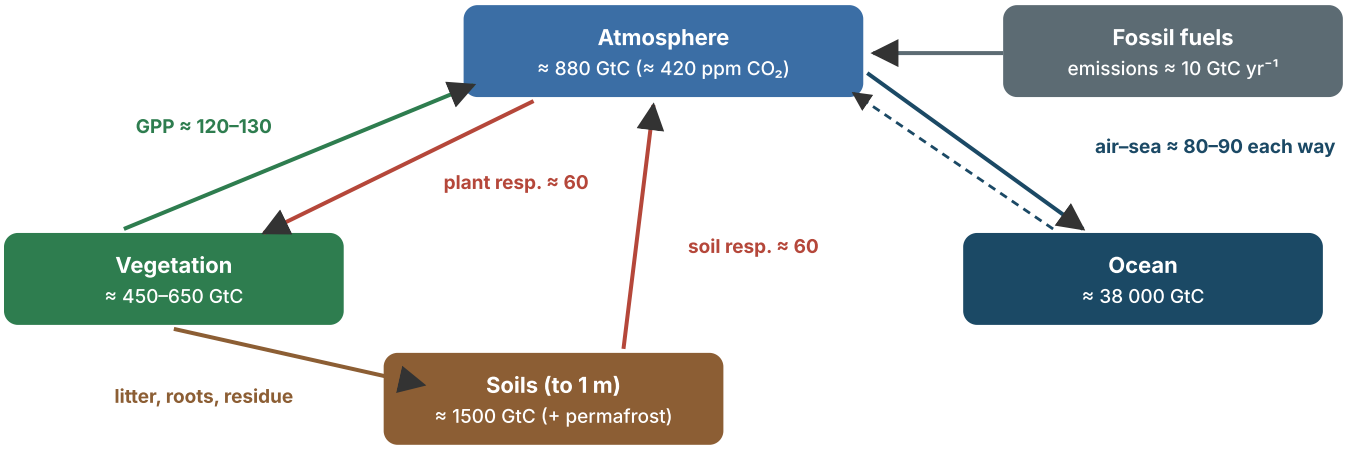
- Salt moves seasonally with water table and irrigation
- Subsurface salinity invisible to optical sensors
- Confusion with sand, bright soils, crop residue
- Needs field EC/ESP calibration

Interview one-liner: Saline soils appear bright with salt crusts and are mapped with indices such as SI, NDSI and brightness index plus vegetation stress, while sodic soils need pH/ESP field data and SWIR or multi-temporal information; dry-season images work best.

Key papers / sources: Metternicht & Zinck (2003) RSE 85:1 · Allbed & Kumar (2013) Adv. Remote Sens. 2:373 · Khan et al. (2005) Agric. Water Manage. 77:96 · Mandal, Sharma et al. (CSSRI) salt-affected soils of India · Richards (1954) USDA Handbook 60

Biogeochemical cycles: carbon cycle and soil carbon change

The carbon cycle moves carbon between atmosphere, vegetation, soils, oceans and fossil reserves. Agriculture affects it through photosynthesis (GPP), respiration, residue and manure inputs, tillage, burning and land-use change. Values below are approximate (Global Carbon Budget).



Main pools (GtC) and annual fluxes (GtC yr^{-1}), approximate

Annual budget (approx., GCB 2023)

- Fossil CO_2 ≈ 10 GtC yr^{-1}; land-use change ≈ 1 GtC yr^{-1}
- Ocean sink ≈ 3; land sink ≈ 3; atmospheric growth ≈ 5 GtC yr^{-1}
- 1 ppm CO_2 ≈ 2.12 GtC; 1 g C = 3.67 g CO_2 (44/12)
- NEP = GPP − R_a − R_h; NEE = −NEP (eddy covariance sign)

Soil organic carbon change

- ΔSOC = inputs (residue, roots, manure) − decomposition (temperature, moisture, tillage)
- SOC stock (t ha^{-1}) = SOC% × BD (g cm^{-3}) × depth (cm) × (1 − coarse fraction)
- Gains from: residue retention, no-till, cover crops, manure, agroforestry, biochar
- Losses: burning, intensive tillage, erosion; '4 per 1000' initiative (COP21)

Measuring / modelling

- Repeated soil sampling (equivalent soil mass)
- Models: RothC, Century/DayCent
- Flux towers (NEE), SIF/GPP, OCO-2 XCO_2
- Biomass: SAR, GEDI lidar; DSM for stocks

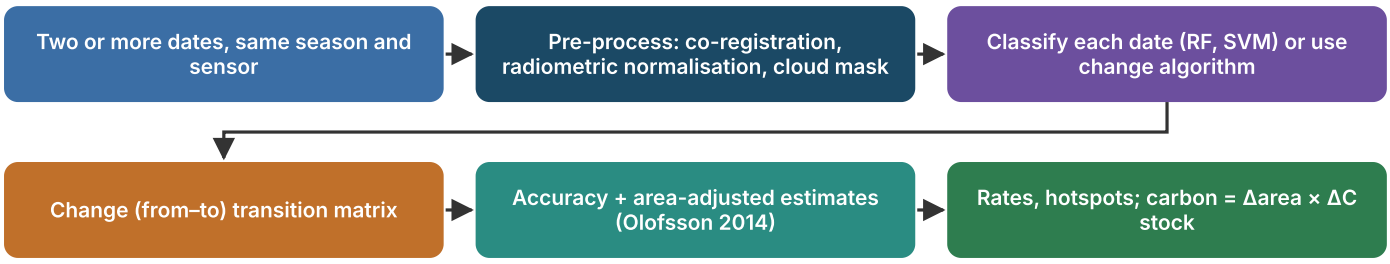
Other cycles	Key agricultural processes
Nitrogen	Fixation (biological, Haber–Bosch), mineralisation, nitrification, denitrification (N_2O), volatilisation (NH_3), leaching (NO_3^-)
Water	Precipitation, infiltration, ET (~60% of land precipitation), runoff, drainage
Phosphorus, sulphur	Weathering, fixation, fertiliser, erosion losses

Interview one-liner: Plants fix about 120 GtC a year by photosynthesis and roughly return it by respiration; soil organic carbon, about 1500 GtC to 1 m, changes with the balance of residue inputs and decomposition, which management such as no-till, residue retention and biochar can shift.

Key papers / sources: Friedlingstein et al. (2023) Earth Syst. Sci. Data 15:5301 (Global Carbon Budget) · Lal (2004) Science 304:1623 · Minasny et al. (2017) Geoderma 292:59 ('4 per 1000') · Coleman & Jenkinson (1996) RothC

Land use / land cover change: detection and quantification

Your notes link the carbon cycle with 'LULC: changes quantified'. Land-use change (deforestation, cropland expansion, urbanisation) emits about 1 GtC a year globally. Remote sensing quantifies where, how much and how fast land cover changes.



LULC change workflow

Method	Idea	Note
Post-classification comparison	Classify each date, compare maps	Gives from–to classes; errors multiply
Image differencing / ratio	Subtract NDVI or bands between dates; threshold	Simple; no class labels
Change vector analysis	Magnitude and direction of spectral change	Detects type of change
PCA / tasseled cap	Change in components (brightness, greenness, wetness)	Multi-band
Time-series algorithms	LandTrendr, BFAST, CCDC on Landsat/Sentinel stacks	Abrupt and gradual change

Quantifying change

- Transition matrix: area moving from class i to j
- Annual rate $r = [1/(t_2 - t_1)]\cdot\ln(A_2/A_1) \times 100$
- Net vs gross change; persistence
- Accuracy: confusion matrix, kappa; error-adjusted area with confidence interval

Indian datasets

- NRSC annual LULC 1:250k (AWiFS, since 2004–05)
- NRSC LULC 1:50k (LISS-III) for 2005–06, 2011–12, 2015–16
- Forest Survey of India biennial forest cover
- Bhuvan LULC services

Carbon link (IPCC)

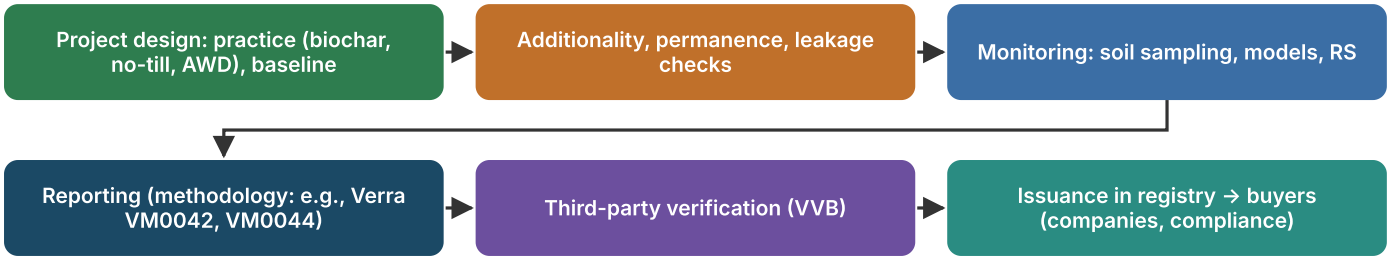
- Stock-difference: $\Delta C = (C_{t2} - C_{t1})/(t_2 - t_1)$
- Gain–loss: emissions = activity data (area) × emission factor
- RS gives activity data; field/biomass maps give factors
- Cropland ↔ forest conversion dominates

Interview one-liner: LULC change is detected by comparing classified maps or spectral differences between dates, summarised in a from–to transition matrix with error-adjusted areas, and converted to carbon emissions by multiplying changed area by the difference in carbon stock.

Key papers / sources: Singh (1989) IJRS 10:989 · Lu et al. (2004) IJRS 25:2365 · Olofsson et al. (2014) RSE 148:42 · Kennedy et al. (2010) RSE 114:2897 (LandTrendr) · Zhu & Woodcock (2014) RSE 144:152 (CCDC) · Verbesselt et al. (2010) RSE 114:106 (BFAST) · NRSC LULC reports

Carbon credits, biochar, regenerative agriculture and MRV

A carbon credit is a tradeable certificate for one tonne of CO_2-equivalent avoided or removed. Agriculture can earn credits by storing carbon in soil or biochar or by cutting emissions (e.g., methane from rice). Credibility depends on MRV: measurement, reporting and verification.



Carbon credit and MRV cycle

Biochar credits

- Pyrolysis of residue at ~350–700 °C with little oxygen
- Stable carbon if $H/C_{org} < 0.7$ (EBC/IBI)
- Credit ≈ biochar C × fraction persisting 100 yr × 44/12 − project emissions
- Alternative to burning; improves soil water and CEC

Regenerative agriculture

- No single definition; outcome-based
- Principles: minimise disturbance, keep soil covered, living roots, diversity, integrate livestock
- Overlaps with CA and natural farming
- Giller et al. (2021): agronomic perspective, claims need evidence

India

- Carbon Credit Trading Scheme 2023 (Energy Conservation Amendment Act 2022); BEE administers Indian Carbon Market
- Offset mechanism can include agriculture
- Green Credit Programme (MoEFCC, 2023)
- Voluntary market: Verra, Gold Standard, Puro.earth

MRV term	Meaning
Baseline	What would happen without the project
Additionality	Reduction would not occur without carbon finance
Permanence	Carbon stays stored (buffer pools for reversal risk)
Leakage	Emissions shifted elsewhere
Remote sensing role	Verify practice (no burning, no-till, cover crops), biomass, area; scale up soil sampling via DSM

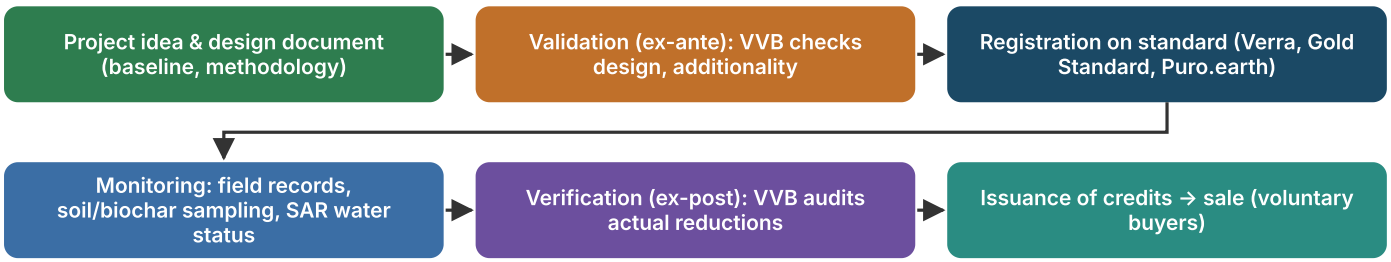
Interview one-liner: A carbon credit equals one tonne CO_2e avoided or removed; biochar and soil-carbon projects must prove baseline, additionality, permanence and no leakage through MRV — measurement (soil sampling, models, remote sensing), reporting under a methodology and independent verification.

Key papers / sources: Smith et al. (2020) Global Change Biol. 26:219 (soil C MRV) · Woolf et al. (2010) Nature Commun. 1:56 (biochar) · Lehmann & Joseph (2015) Biochar for Environmental Management · Giller et al. (2021) Outlook Agric. 50:13 · Verra VM0042 / VM0044 · BEE CCTS (2023)

Rice methane, DSR and AWD; voluntary carbon credits and their validation

Your notes list regenerative options for rice — moving from puddled transplanting to direct-seeded rice (DSR) and alternate wetting and drying (AWD) — and ask how voluntary carbon credits are validated, including biochar incorporation. Flooded rice soils are anaerobic, so methanogens produce methane (CH_4, $GWP_{100} \approx 27–30$).

Practice	How it works	Benefits	Trade-offs
Puddled transplanted rice (PTR)	Wet tillage + continuous flooding	Weed control, easy transplanting	High water use, CH_4, soil hard pan, labour
Direct-seeded rice (DSR)	Seed sown directly (dry or wet); no puddling or nursery	Saves water and labour; earlier harvest; less CH_4	Weeds, herbicide needs, possible yield loss if poorly managed
AWD (alternate wetting and drying)	Irrigate when water in a field tube falls ~15 cm below surface ('safe AWD')	Large CH_4 cut and water saving	Slight N_2O rise; needs water control
Residue + biochar	Straw pyrolysed to biochar and incorporated	Stable C storage; avoids burning	Cost, logistics



Voluntary carbon credit cycle: validation of design vs verification of results

Validating biochar credits

- Feedstock and production records (temperature, yield)
- Lab tests: C content, $H/C_{org} < 0.7$, contaminants (EBC/IBI)
- Chain of custody and application records (where, how much incorporated)
- Permanence factor (fraction stable at 100 yr); methodologies: Verra VM0044, Puro.earth, EBC C-Sink

Remote sensing for rice MRV

- Sentinel-1 SAR: flooded fields have very low backscatter; drying events show as rises → AWD verification
- DSR detection: no puddling/flooding phase before sowing
- Rice area and calendar for baseline emissions
- Methodologies: CDM AMS-III.AU, Gold Standard rice water management, Verra VM0051 (rice)

Interview one-liner: DSR avoids puddling and AWD lets fields dry periodically, both cutting methane from anaerobic rice soils; for voluntary credits, validation checks the project design before it starts and verification audits the measured reductions afterwards, with SAR able to verify flooding and drying cycles.

Key papers / sources: IRRI safe-AWD guidelines · Lampayan et al. (2015) Field Crops Res. 170:95 (AWD) · Kumar & Ladha (2011) Adv. Agron. 111:297 (DSR) · Verra VM0044 · CDM AMS-III.AU · IPCC AR6 WG1 Ch. 7 (GWP)

Remote-sensing-based government programmes: soil and land

Many national programmes use satellite data and GIS to map soils, degraded land and watersheds. Know the agency, scale and what remote sensing contributes.

Programme / product	Agency	Remote sensing / GIS role
Soil resource mapping; Land Resource Inventory (LRI)	ICAR-NBSS&LUP	Soil maps 1:250k → 1:50k; LRI at 1:10k using high-resolution imagery + DEM for landforms and soil-site units
Soil Health Card scheme (2015)	MoA&FW	Grid sampling of soils; geo-tagged samples, GIS portal; nutrient maps (not satellite-based testing)
Digital Agriculture Mission (2024)	MoA&FW	Krishi Decision Support System (geospatial), digital crop survey, detailed soil profile mapping (~1:10k)
Desertification & Land Degradation Atlas	SAC-ISRO	Multi-date IRS (AWiFS/LISS-III) mapping of degradation processes; 2018–19: ~97.8 Mha (~29.7% of India) degraded
Wastelands Atlas of India	DoLR + NRSC	Multi-temporal IRS data, 1:50k, several editions
Salt-affected soils of India	NRSC, CSSRI, NBSS&LUP	Satellite-based mapping (~6.7 Mha) with field EC/pH
Watershed Development (PMKSY-WDC, earlier IWMP)	DoLR + NRSC	Bhuvan-based monitoring: pre/post satellite comparison, Srishti/Drishti geotagging apps
Geo-MGNREGA	MoRD + NRSC	Geotagging of assets (farm ponds, check dams) on Bhuvan
Soil moisture products	NRSC (hydrological model), MOSDAC (SAC)	Daily soil moisture/ET for drought and water management
Groundwater prospects maps	NRSC (national drinking water mission)	Hydrogeomorphology from satellite at 1:50k

How RS helps soil programmes

- Landform and relief from DEM (CartoDEM) → soil-site units
- Bare-soil composites for texture/SOC (DSM)
- Degradation: erosion, salinity, waterlogging, vegetation loss
- Monitoring watershed treatment impact (NDVI, water bodies)

Platforms

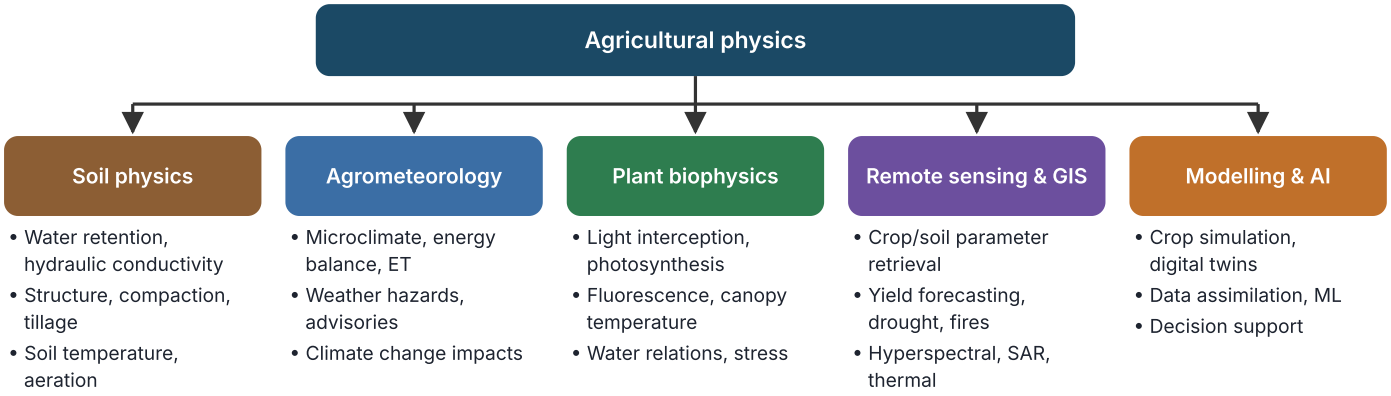
- Bhuvan (NRSC): thematic layers, mobile geotagging
- MOSDAC (SAC): meteorology, soil moisture, LST
- VEDAS (SAC): visualisation and analytics
- Krishi-DSS: national agricultural geospatial platform

Interview one-liner: NBSS&LUP's land resource inventory at 1:10,000, SAC's desertification atlas, NRSC's wasteland and salt-affected soil maps, and Bhuvan-based watershed and MGNREGA monitoring are the main satellite-based soil and land programmes; the Digital Agriculture Mission adds a geospatial decision support system and detailed soil profile maps.

Key papers / sources: SAC-ISRO Desertification & Land Degradation Atlas of India (2021) · NBSS&LUP Land Resource Inventory reports · DoLR/NRSC Wastelands Atlas · PIB notes on Digital Agriculture Mission (Sept 2024) · NRSC Bhuvan documentation

Agricultural physics: definition, objectives, achievements and IARI soil values

Agricultural physics applies the principles of physics to the soil–plant–atmosphere system to understand and manage the transfer of energy, water, heat, gases and momentum for better crop production and resource conservation. Keep this answer to one or two lines in the interview.



Objectives (division)

- Characterise physical environment of crops (soil, weather)
- Develop technologies for water, energy and soil conservation
- Remote sensing and modelling for crop monitoring and forecasting
- Climate risk management and agromet services
- Teaching and capacity building (M.Sc., Ph.D.)

Achievements to quote (verify locally)

- Soil physical characterisation, pedotransfer functions, mulching/tillage practices
- Remote sensing yield and crop condition methods contributing to FASAL-type work
- CREAMS: satellite ground station, daily residue-fire bulletins
- Agromet advisories (Agromet Field Unit, IARI)
- Hyperspectral libraries and RTM studies (wheat LAI, chlorophyll)

IARI farm soil (typical published values)	Value
Taxonomy / texture	Typic Haplustept (Inceptisol); sandy loam
Bulk density	≈ 1.45–1.60 Mg m^{-3} (surface)
Field capacity (−33 kPa)	≈ 0.20–0.25 m^3 m^{-3}
Permanent wilting point (−1500 kPa)	≈ 0.07–0.09 m^3 m^{-3}
Available water	≈ 130–160 mm per m depth
pH; organic C	≈ 7.5–8.2; ≈ 0.4–0.5%

The IARI values are typical ranges reported in published field experiments, not official division figures; replace them with your own measured values before the interview (your notes ask for experimental values of BD, texture and WHC).

Interview one-liner: Agricultural physics applies physics to the soil–plant–atmosphere continuum — soil physics, agrometeorology, plant biophysics, remote sensing and modelling — to manage water, energy and crop growth.

Key papers / sources: IARI Division of Agricultural Physics pages · Hillel (2004) Introduction to Environmental Soil Physics · Monteith & Unsworth (2013) Principles of Environmental Physics

Horticultural vs field crop models

Field crop models simulate one annual cycle where yield ≈ biomass × harvest index. Horticultural models (fruit trees, vegetables, plantation crops) must also handle perennial structure, dormancy, bearing habit, fruit quality and intensive management, so a wheat model cannot simply be reused for an orchard.

Aspect	Field crop model	Horticultural crop model
Life cycle	Annual; reset each season	Perennial: reserves, wood and structure carry over; vegetables with multiple harvests
State variables	LAI, biomass, phenology, soil water and N	+ tree age, canopy volume, wood biomass, carbohydrate reserves, fruit number and size
Phenology	Thermal time and photoperiod	Chilling requirement, dormancy break, bud burst, flowering induction, fruit set
Growth	RUE or photosynthesis; fixed partitioning	Carbon balance with source–sink competition; fruit is a strong sink
Yield	Biomass × harvest index	Fruit number × fruit size + quality (sugar, size grade, colour, shelf life)
Canopy	Uniform 'big leaf'	Hedgerow or individual trees; 3-D light interception
Management	Sowing, irrigation, N	Pruning, training, thinning, girdling, fertigation, growth regulators, protected cultivation
Special problems	Water/N stress, lodging	Alternate bearing, flower/fruit drop, frost at bloom, cracking, sunburn
Validation	Multi-location trials in one season	Multi-year orchard data needed (carry-over effects)



Perennial fruit-tree model logic

Model examples (verify names)

- Field: DSSAT (CERES, CROPGRO), APSIM, WOFOST, InfoCrop (IARI), AquaCrop (FAO)
- Vegetable/greenhouse: TOMGRO, CROPGRO-tomato, HORTSYST
- Fruit/perennial: STICS (grapevine, fruit trees), VineLOGIC, L-PEACH, MaluSim (apple)
- Few process models for mango, banana, citrus → RS + statistical models used

Remote sensing and ISRO

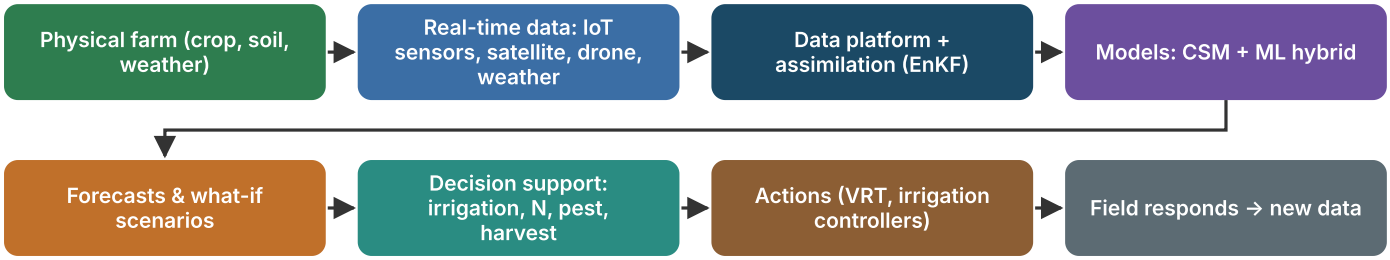
- CHAMAN: orchard mapping, area, production, site suitability
- Horticulture harder: small, fragmented, perennial, spectrally similar
- RS gives area, age, canopy (LAI, chlorophyll), stress, yield indices
- Challenges: few long datasets, cultivar parameters, management complexity

Interview one-liner: Field crop models simulate one annual cycle driven by biomass and harvest index, whereas horticultural models must represent perennial reserves, chilling and dormancy, source–sink fruit competition, canopy management and fruit quality; with few process models, programmes like CHAMAN rely heavily on remote sensing.

Key papers / sources: Jones et al. (2003) Eur. J. Agron. 18:235 (DSSAT) · Keating et al. (2003) Eur. J. Agron. 18:267 (APSIM) · Aggarwal et al. (2006) Agric. Syst. 89:1 (InfoCrop) · Brisson et al. (2003) Eur. J. Agron. 18:309 (STICS) · Allen et al. (2005) Ecol. Model. (L-PEACH) · Richardson et al. (1974) HortScience 9:331 (chill units)

Crop simulation models vs digital twins

A crop simulation model (CSM) mathematically simulates daily crop growth from weather, soil, cultivar and management. A digital twin is a living virtual replica of a real farm that is continuously updated with real-time data (sensors, satellites, drones) and feeds decisions back to the field. CSMs often sit inside a digital twin as its physics engine.



Digital twin loop (bi-directional, real time)

Aspect	Crop simulation model	Digital twin
Purpose	Understand and predict crop growth; scenario analysis	Monitor, predict and control a specific real system
Data link	Offline inputs (weather files, soil, management)	Continuous, real-time data streams
Updating	Calibrated once; run forward	State updated by data assimilation; learns over time
Direction	One-way (model → output)	Two-way (field ↔ model; actions feed back)
Scale	Point/field, generic	Specific farm, plot or plant; spatially explicit
Core	Process equations (phenology, RUE, water, N balance)	CSM + ML + IoT + RS + decision engine
Examples	DSSAT, APSIM, WOFOST, InfoCrop, AquaCrop, ORYZA	Research prototypes; smart greenhouse twins

What a CSM needs

- Weather: daily Tmax, Tmin, rainfall, radiation
- Soil: layers, water limits, N
- Cultivar (genetic) coefficients; management
- Outputs: phenology, LAI, biomass, yield, water/N balance

Linking with remote sensing

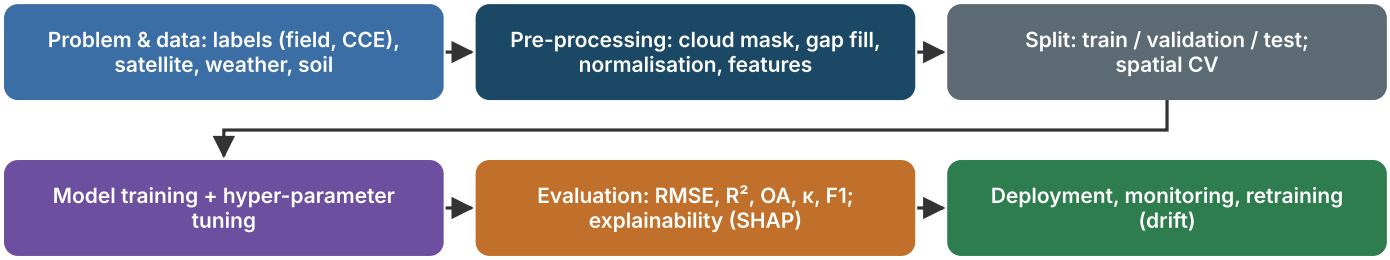
- Forcing: satellite weather (rainfall, radiation)
- Calibration: LAI, phenology from satellite
- Assimilation: update LAI/soil moisture (EnKF) → better yield (YES-TECH, FASAL)
- Upscaling field model to region with RS crop maps

Interview one-liner: A crop simulation model predicts crop growth from weather, soil and management in one direction, while a digital twin wraps such models in a real-time, two-way loop with sensors, satellites and assimilation so the virtual farm stays synchronised with the real one and drives decisions.

Key papers / sources: Pylianidis, Osinga & Athanasiadis (2021) Comput. Electron. Agric. 184:105942 · Verdouw et al. (2021) Agric. Syst. 189:103046 · Jones et al. (2003) · Huang et al. (2019) Agric. For. Meteorol. 276:107609 (assimilation into crop models)

Machine learning and AI: core concepts for agriculture

Machine learning (ML) learns patterns from data instead of explicit rules; deep learning uses many-layered neural networks; AI is the broader goal of machines performing intelligent tasks. In agriculture ML maps crops, predicts yield, detects pests and downscales weather, usually combined with remote sensing.



ML workflow

Family	Algorithms	Typical agricultural use
Supervised regression	Linear/ridge/lasso, random forest (RF), SVR, gradient boosting (XGBoost), GPR	Yield, LAI, biomass, soil properties (DSM)
Supervised classification	RF, SVM, k-NN, maximum likelihood, decision trees	Crop type mapping, land cover, burnt area
Unsupervised	k-means, ISODATA, PCA, clustering	Management zones, exploratory mapping, dimensionality reduction
Deep learning	CNN (images), U-Net (segmentation), LSTM/GRU (time series), transformers	Field boundaries, pest/disease images, SAR/optical time series
Hybrid / physics-informed	ML trained on RTM or crop model simulations; ML emulators	Operational LAI, YES-TECH ensembles, digital twins
Generative / LLM	Foundation models, chatbots	Farmer advisories in local languages

Key concepts

- Overfitting vs underfitting; bias–variance trade-off
- Regularisation, dropout, early stopping
- Cross-validation (k-fold; spatial/temporal blocks to avoid leakage)
- Feature importance, SHAP for explainability
- Class imbalance, data augmentation, transfer learning

Metrics

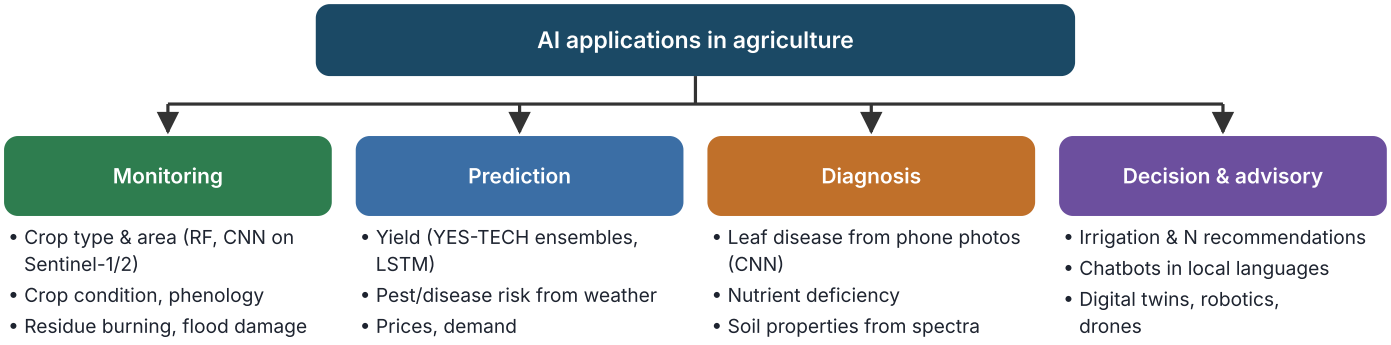
- Regression: RMSE, MAE, bias, R^2, nRMSE, NSE
- Classification: confusion matrix, overall accuracy, producer's & user's accuracy, kappa, F1
- Area: pixel counting vs regression estimator; sampling-based area estimate
- Uncertainty: prediction intervals, ensembles

Interview one-liner: Supervised ML such as random forest learns crop or yield from labelled field data and satellite features; deep learning (CNN, LSTM) handles images and time series; models must be validated with spatially independent data and evaluated with RMSE, R^2, accuracy and kappa.

Key papers / sources: Breiman (2001) Mach. Learn. 45:5 (random forest) · LeCun, Bengio & Hinton (2015) Nature 521:436 · Liakos et al. (2018) Sensors 18:2674 (ML in agriculture review) · van Klompenburg et al. (2020) Comput. Electron. Agric. 177:105709 (yield prediction review) · Belgiu & Drăguţ (2016) ISPRS J. 114:24 (RF in RS)

AI in Indian agriculture and the IAIRO

AI applications in agriculture combine sensing (satellite, drone, phone camera, IoT) with ML models and decision tools. India is building national AI capacity through the IndiaAI Mission and the new Indian AI Research Organisation (IAIRO).



IAIRO (Indian AI Research Organisation)

- Launched January 2026; GIFT City, Gandhinagar
- Section 8 non-profit; public–private partnership (Gujarat, Centre under IndiaAI Mission, Indian Pharmaceutical Alliance)
- ₹300 crore over five years
- Aim: 'ISRO for AI' — sovereign, applied AI (health, education, climate), IP and start-ups

National context

- IndiaAI Mission (March 2024, ≈ ₹10,372 crore): compute, datasets, foundation models
- Digital Agriculture Mission, AgriStack (farmer and land registries)
- Kisan e-Mitra and other AI advisory pilots
- ICAR-NePPA: AI, sensors, drones in precision agriculture

Issues to discuss

- Data quality and labels (CCEs, field surveys)
- Small fields and mixed pixels
- Explainability and trust for policy use (insurance)
- Bias across regions; model drift with climate

Physics-based vs ML	Strength	Weakness
Physics (RTM, CSM, SEB)	Generalises, interpretable, needs less data	Needs parameters; simplifications
ML	Accurate with enough data; handles complex relations	Data hungry; may not transfer; black box
Hybrid (best practice)	Physics constraints + ML flexibility	More complex to build

Interview one-liner: AI in agriculture links sensing with ML for monitoring, prediction, diagnosis and advice; India's IAIRO, launched in January 2026 at GIFT City with ₹300 crore as a public–private partnership under the IndiaAI Mission, aims to be an 'ISRO for AI' focused on applied, sovereign AI.

Key papers / sources: Business Today / The Week reports on IAIRO launch (30 Jan 2026) · Analytics India Magazine 'Inside IAIRO' · IndiaAI Mission (MeitY, 2024) · Liakos et al. (2018) Sensors 18:2674 · Kamilaris & Prenafeta-Boldú (2018) Comput. Electron. Agric. 147:70 (deep learning in agriculture)

Basics revision: soil physics and meteorology

Your notes list 'basics: soil physics, meteorology, statistics and ML'. These are the formulas and numbers most often asked.

Soil physics	Formula / value
Bulk density, porosity	$\rho_b = M_s/V_t$; $f = 1 - \rho_b/\rho_s$ ($\rho_s \approx 2.65$ Mg m^{-3})
Water content	$\theta_v = \theta_g \times \rho_b/\rho_w$; depth of water = $\theta_v \times$ soil depth
Texture (USDA)	Sand 2–0.05 mm, silt 0.05–0.002 mm, clay < 0.002 mm
Water limits	FC ≈ −33 kPa (−10 kPa sandy), PWP = −1500 kPa; AWC = FC − PWP
Darcy's law	$q = -K\,\partial H/\partial z$ (saturated); Buckingham–Darcy for unsaturated $K(\theta)$
Richards equation	$\partial\theta/\partial t = \partial/\partial z[K(h)(\partial h/\partial z + 1)]$
van Genuchten	$\theta(h) = \theta_r + (\theta_s - \theta_r)/[1 + (\alpha\lvert h\rvert)^n]^{1-1/n}$
Infiltration	Philip: $I = S\,t^{1/2} + A\,t$; Green–Ampt; Kostiakov
Soil heat	$G = -\lambda\,\partial T/\partial z$; damping depth $D = \sqrt{(2\kappa/\omega)}$

Meteorology	Formula / value
Radiation laws	Stefan–Boltzmann $E = \varepsilon\sigma T^4$ ($\sigma = 5.67\times10^{-8}$); Wien $\lambda_{max} = 2898/T$ µm; solar constant ≈ 1361 W m^{-2}
Planck / brightness T	$BT = K_2 / \ln(K_1/L + 1)$; Landsat 10: $K_1 = 774.89$, $K_2 = 1321.08$
Vapour pressure	$e° = 0.6108\exp(17.27T/(T + 237.3))$ kPa; VPD = $e_s - e_a$
Energy balance	$Rn = H + \lambda E + G$; $\beta = H/\lambda E$; $\lambda \approx 2.45$ MJ kg^{-1}; 1 mm d^{-1} ≈ 28.4 W m^{-2}
Lapse rates	Dry 9.8 K km^{-1}; standard 6.5 K km^{-1}
Crop weather	$GDD = \Sigma[(T_{max} + T_{min})/2 - T_{base}]$; PAR ≈ 0.48 Rs; biomass = RUE × ΣAPAR
Wind profile	$u(z) = (u^*/k)\ln[(z - d)/z_0]$

Quick numbers

- Monsoon LPA 87 cm (1971–2020)
- India salt-affected ≈ 6.7 Mha; degraded ≈ 97.8 Mha (2018–19)
- Average holding ≈ 1.08 ha
- Atmosphere ≈ 420 ppm CO_2 ≈ 880 GtC

Remote sensing basics

- Resolution: spatial, spectral, radiometric, temporal
- NDVI = (NIR − R)/(NIR + R); EVI; SAVI (L = 0.5)
- SAR: VV/VH backscatter, incidence angle, speckle
- Thermal: emissivity, split window, LST

Basics revision: statistics for remote sensing and modelling

Most interview questions on statistics are about how you validated a model or estimated an area.

Statistic	Formula	Use
Mean, SD, CV	$\bar{x}$; $s = \sqrt{[\Sigma(x - \bar{x})^2/(n-1)]}$; CV = $s/\bar{x} \times 100$	Variability of yield, soil properties
Correlation r	$\text{cov}(x, y)/(s_x s_y)$	Linear association
R^2	$1 - SS_{res}/SS_{tot}$	Variance explained (not accuracy)
RMSE, nRMSE, MAE, bias	$\sqrt{[\Sigma(P - O)^2/n]}$; RMSE/$\bar{O}$; $\Sigma\lvert P - O\rvert/n$; $\Sigma(P - O)/n$	Model error
Nash–Sutcliffe (NSE)	$1 - \Sigma(O - P)^2/\Sigma(O - \bar{O})^2$	Hydrology, crop models (1 = perfect)
Willmott d	$1 - \Sigma(P - O)^2/\Sigma(\lvert P - \bar{O}\rvert + \lvert O - \bar{O}\rvert)^2$	Agreement index
Confusion matrix	OA = Σdiagonal/N; producer's (omission), user's (commission) accuracy	Crop classification
Kappa κ	$(p_o - p_e)/(1 - p_e)$	Agreement beyond chance
t-test, ANOVA	Compare means of treatments	Field experiments
Regression	$y = a + bx + \varepsilon$; multiple, stepwise; check residuals	Agromet yield models, VI models

Sampling

- Simple random, stratified random (FASAL area), systematic, cluster
- Conditioned Latin hypercube (cLHS) for soil
- Smart sampling of CCEs using RS strata
- Sample size ∝ $(Z \cdot CV/E)^2$

Geostatistics

- Semivariogram $\gamma(h) = \frac{1}{2} \cdot \text{mean}[z(x) - z(x + h)]^2$: nugget, sill, range
- Kriging (ordinary, universal, co-kriging) = best linear unbiased estimator
- IDW, splines as deterministic alternatives
- Regression kriging in DSM

Rapid-fire interview questions with one-line answers

Practise answering each in one or two sentences, then expanding if asked.

Question	Short answer
How is daily ET obtained from a satellite snapshot?	Keep EF (or ETrF = ET/ETr) constant through the day and multiply by daily Rn (or daily reference ET).
SEBAL vs METRIC vs ALEXI?	SEBAL/METRIC single-source with hot/cold pixels (METRIC ties cold pixel to 1.05 ETr); ALEXI two-source, uses morning LST rise, separates E and T.
Why SAR for kharif?	Clouds block optical sensors in monsoon; SAR sees through cloud.
What is YES-TECH?	Technology-based yield estimation under PMFBY, 30% weight with CCE yield for paddy and wheat.
What is smart sampling?	Using satellite crop-vigour strata to place fewer, more representative crop-cutting experiments.
How is drought declared?	Rainfall trigger, then ≥ 3 of 4 impact indicators (crop, RS, soil moisture, hydrology), then ground truthing.
What does FLEX measure?	Sun-induced chlorophyll fluorescence (FLORIS, 300 m), launched Sept 2026 with Sentinel-3C.
What is NISAR's band and repeat?	L- and S-band SAR, 12-day repeat; crop mapping and ~200 m soil moisture.
How are residue fires monitored?	VIIRS/MODIS active fires and FRP (CREAMS daily bulletin); Sentinel-2 NBR for burnt area.
Index for residue/CA?	NDTI = (SWIR1 − SWIR2)/(SWIR1 + SWIR2) or CAI (2.0, 2.1, 2.2 µm).
What is MRV?	Measurement, reporting and verification of carbon gains for credits.
Obukhov length sign?	Negative unstable (day), positive stable (night), very large neutral.
Richardson number critical value?	About 0.25; above it turbulence is suppressed.
Eddy covariance flux?	30-min covariance of vertical wind with T, q or CO_2 at 10–20 Hz.
CSM vs digital twin?	CSM simulates offline; digital twin is a real-time, two-way, data-assimilating replica.
Why not a wheat model for mango?	Perennial reserves, chilling/dormancy, alternate bearing and fruit quality need different processes.
What is CA?	Minimum soil disturbance + permanent soil cover + crop diversification (FAO).
What is IAIRO?	Indian AI Research Organisation, GIFT City, launched Jan 2026, ₹300 crore PPP, 'ISRO for AI'.

Key papers and sources

All papers cited on the topic pages, alphabetical. Journal abbreviations: RSE = Remote Sensing of Environment, GRL = Geophysical Research Letters, JGR = Journal of Geophysical Research, HESS = Hydrology and Earth System Sciences.

- Aggarwal et al. (2006) Agric. Syst. 89:1 (InfoCrop)
- Allbed & Kumar (2013) Adv. Remote Sens. 2:373
- Allen et al. (2005) Ecol. Model. (L-PEACH)
- Allen, Pereira, Raes & Smith (1998) FAO Irrigation & Drainage Paper 56
- Allen, Tasumi & Trezza (2007) J. Irrig. Drain. Eng. 133:380
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